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Recent advances in the development of stereolithography-based additive manufacturing processes: A review of applications and challenges

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ABSTRACT

Additive manufacturing processes have progressed over recent years due to their superiority over conventional manufacturing methods. Their ability to fabricate materials with complex structures, increased precision, and reduced cost have opened avenues for various industrial applications, including biomedical, electrical, mechanical, aviation, and filtration, and led to their development over time. Stereolithography (SLA) is an additive manufacturing technique, through photopolymerization reaction, it solidifies a selective resin to produce three-dimensional objects. SLA has emerged as a leading 3D printing technique, revolutionizing prototyping and production across various industries. SLA has been through four generations of development and advancement, resulting in its improved performance, the diversity of materials, and the variety of applications. Stereolithography has diversified its material and emerged as a promising method for polymer-based composite when operating under optimized conditions. SLA offers superior resolution, high finish quality, improved speed and precision, and is cost-effective compared to alternative techniques like Fused Deposition Modeling (FDM). This current study aims to comprehensively review SLA development, its processes, applications and inherent challenges in mechanical, electrical and biomedical fields.

1. Introduction

Additive manufacturing (AM) or 3D printing (3DP) processes have many advantages over traditional manufacturing processes due to their ability to fabricate complex 3D structures with a wide range of materials [1]. 3DP uses computer-aided designs to construct a three-dimensional structure by layering the material. They have the potential to produce intricate components with advanced features in functionality, geometry, and material composition [2,3]. Due to continuous development and improvement over time, new materials are being employed for production, which has widened the application of these processes in the aerospace, filtration, biomedical, mechanical, and electrical fields [4]. Based on the American Society for Testing and Materials (ISO/ASTM 52900:2015), AM processes are divided into seven categories depending upon their operational principles [5]. These categories include vat polymerization, material extrusion, material jetting, binder jetting, powder bed fusion, direct energy deposition, and sheet lamination [6,7]. The selection of the AM process is primarily influenced by the desired properties of the final product, the specific material used, and the complexity of the design [8]. Vat polymerization is the first ever developed process that utilizes a light source to trigger a polymerization reaction, which then cures photo-sensitive material layer by layer, resulting in a 3D structure [9,10]. Based on the light source and polymerization mechanism employed, these processes are further grouped into stereolithography (SLA), two-photon polymerization (2 PP), digital light polymerization (DLP), and volumetric 3D printing [11,12].

Stereolithography (SLA) is the earliest vat polymerization technique, developed by Charles Hull in 1986 [13,14]. It is based on the principle of photopolymerization and utilizes a laser source (UV light) to trigger polymerization reactions in photo-sensitive material [15,16]. The photopolymer resin consists of precursors (monomers or oligomers), binders, photoinitiators (PI), absorbers, additives and chain transfer agents [17]. The precursors are the main element of the resin that solidifies due to cross-linking [18]. The photoinitiators transform into radicals when exposed to light, which aids cross-linking between the monomers/oligomers, resulting in polymeric chains [19,20]. Furthermore, additives in the form of nanoparticles and colours are incorporated to improve properties or customize the printed object [21,22]. Due to being a single photon reaction, the process occurs on the surface of the resin, resulting in layer-by-layer structure formation [10]. Precision and

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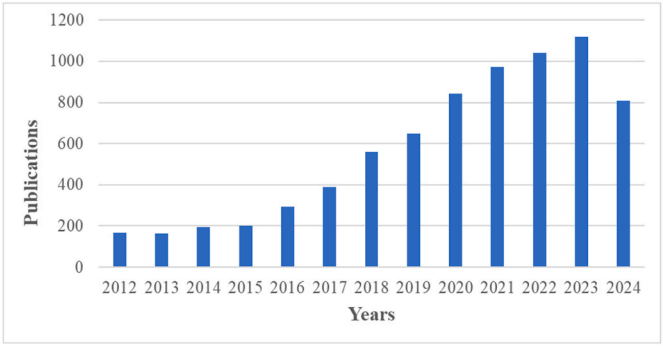


Fig. 1. Publication hits for "Stereolithography (SLA)" data from Scopus, accessed September 1st, 2024.

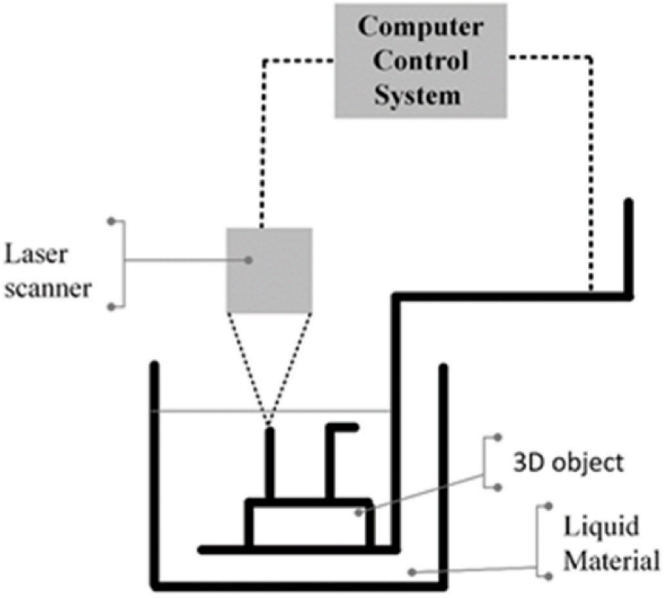


Fig. 2. Stereolithography Working Principle (Adapted from Ref. [32] Coprights © 2020 by the authors. Licensee MDPI, Basel, Switzerland).

accuracy of the light source substantially affect the final product, so Galvano scanners are used to maintain the accuracy and printing speed of the laser [6]. Stereolithography has evolved significantly beyond its initial use of prototyping and modelling. It is expanding its capabilities in manufacturing highly complex and intricate designs [1,23]. The range of materials has evolved from traditional polymers to composites, metals, and ceramics. This enhanced material versatility has provided opportunities for creating products tailored for industries like aerospace, robotics, sensors [24], medical implants [25–27], energy storage components [28–30], microfluidic devices [31] and more [32]. Nonetheless, it is essential to note that SLA primarily involves structuring a single material at a time. The kinetics of the reaction and curing process are also intricate and multifaceted [20]. Compared to other additive manufacturing (AM) techniques, SLA provides superior resolution (as low as 10 μm), enhanced surface quality and higher printing speed, albeit at the trade-off of higher manufacturing cost [21,33]. SLA offers significant economic and practical benefits beyond its technical capabilities. Rapid prototyping, adaptability, customization and material versatility make it much more cost-effective and economically viable than traditional manufacturing techniques. In a study by Yang et al. [34], an optimization problem for SLA was developed to lower the costs of additive manufacturing while taking into account the set of decision variables (surface stratification angle and layer thickness) and adhering to production throughput and part quality limitations. They revealed that by resolving the suggested optimization challenge, cost savings of 26 % can be achieved. Another study by Nagarajan et al. [35] performed an environmental impact assessment (EIA) for a novel fast mask image projection stereolithography (MIP-SL) process used to fabricate six functional test parts. This assessment documents the materials, energy, and waste associated with the process. The assessment revealed the relationship between energy use and environmental impact, facilitating predictions of process energy consumption through a mathematical model that correlates shape complexity and part dimensions with build and washing times so that process energy consumption can be controlled at the part design stage. SLA is also known to enable design freedom, reduce waste production, integrate with other processes, and ensure quality control. By leveraging these benefits, organizations can accelerate product development, improve efficiency, and drive innovation. With ongoing advancements in technology and user-friendly interfaces, SLA provides a balance of performance and cost-efficiency, making it a practical and economically viable option for both prototyping and production.

From 2012 to 2024, the field of stereolithography has experienced substantial growth in research activity, reflecting its expanding applications and technological advancements. From 2012 to 2024, the field of SLA has seen significant growth in research activity, marked by approximately 7400 publications in the Scopus database (Fig. 1). This surge in publications indicates heightened interest and research activity driven by technological advancements, broader applications, and greater accessibility. With the increasing volume of literature on stereolithography technology and its application, it is crucial to condense the knowledge into a concise format, which enables researchers to grasp vital information and identify research gaps and avenues. Additionally, they can have comprehensive knowledge of the materials, processes and applications and can pinpoint the emerging trends and future trajectories in SLA technology as well as confront the challenges they face along the way, thereby establishing a foundational framework for innovations and progress in the field. This article provides a comprehensive overview of stereolithography processes and systems. It categorizes and highlights the development of different stereolithography systems based on the processes. Furthermore, it details the processes involved in various applications, the material used, and the challenges faced.

1.1. Methodology

A comprehensive review of peer-reviewed articles, review papers, books, magazines, and internationally reputed blogs was conducted through different search databases, including Google Scholar, Springer, Scopus, Wiley, PubMed, and ProQuest. The literature search utilized keywords in the articles and the abstract, including terms related to stereolithography, 3D Printing, additive manufacturing, polymers, stereolithographic techniques, mechanical, electrical and biomedical applications, and the combination of these keywords. The primary source for the literature review was research or review articles, conference proceedings, and books were also considered. Subsequently, the articles were subjected to detailed examination to extract information based on stereolithography principles, materials, process parameters, applications, prospects and challenges. The extracted data were then thematically organized into stereolithography development, resin chemistry and application in biomedical, mechanical and electrical fields.

2. Stereolithography (SLA)

SLA employs UV light to initiate a chain polymerization reaction within a monomer solution and selectively solidify each layer [36]. The UV active monomer reacts with other monomers in the resin and transforms into a polymer chain. Following polymerization, one resin layer solidifies, providing a base for the subsequent layers, forming a 3D

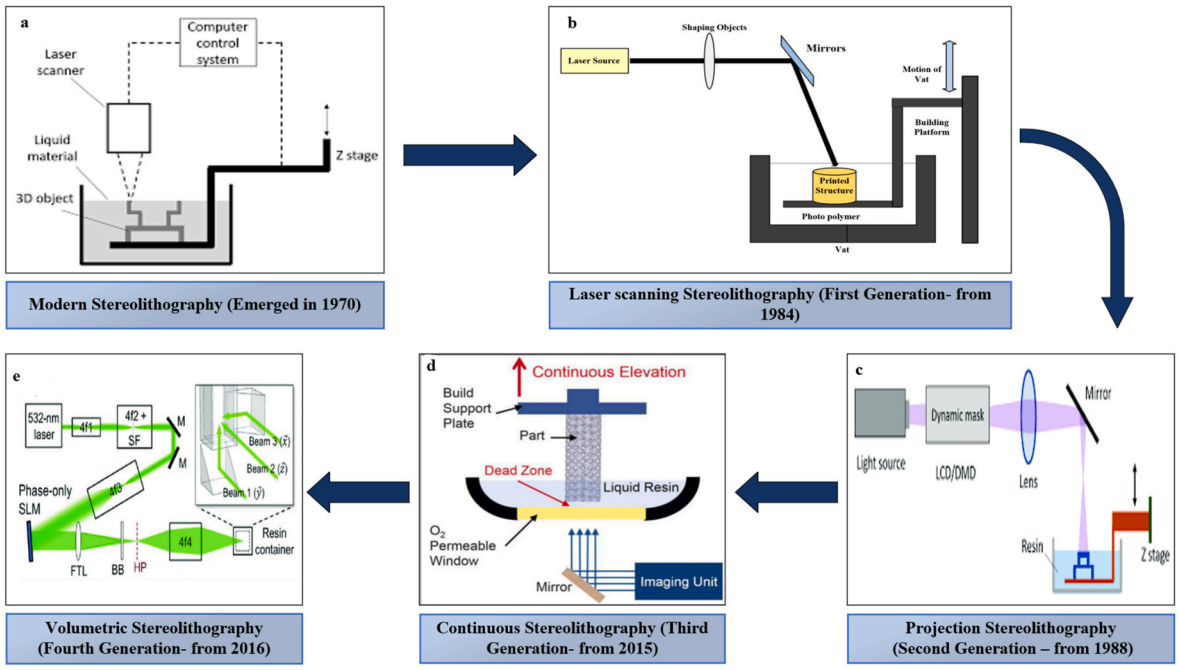


Fig. 3. Four Generations of Stereolithography. (a) Modern Stereolithography (Adapted from Ref. [32] Coprights © 2020 by the authors. Licensee MDPI, Basel, Switzerland), (b) Laser Scanning Stereolithography, (c) Projection Stereolithography **(**Adapted from Ref. [32] Coprights © 2020 by the authors. Licensee MDPI, Basel, Switzerland), (**d**) Contionous Stereolithography (Adapted from Ref. [44] Copyrights © 1999–2024 John Wiley & Sons, Inc), (e) Volumetric Stereolithography (Adapted from Ref. [45] Copyrights © 2017 American Association for the Advancement of Science).

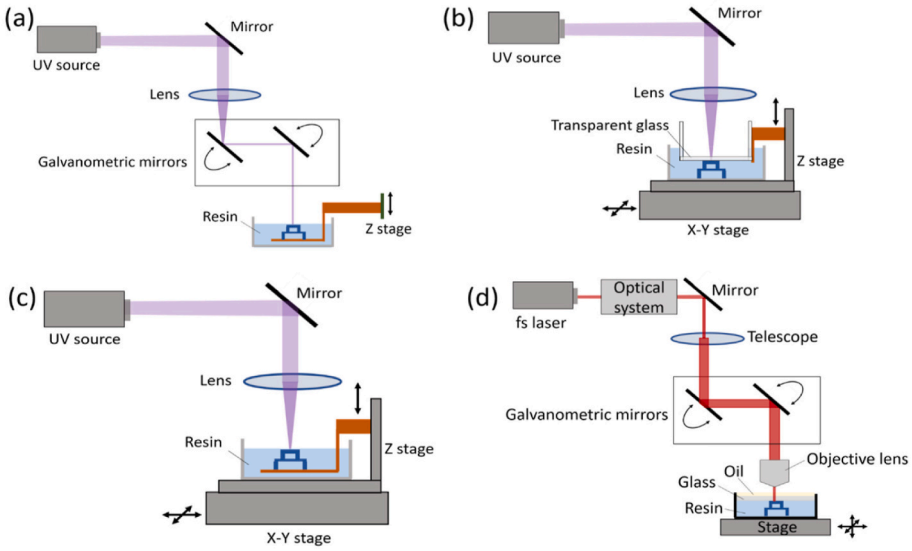


Fig. 4. Laser-Scanning Stereolithography **(a)** shows Dynamic Light Scanning stereolithography **(b)** shows the constraint-surface system **(c)** The free-surface system **(d)** Two-photon Polymerization (2 PP) system (Adapted from Ref. [32] Coprights © 2020 by the authors. Licensee MDPI, Basel, Switzerland).

structure [36,37]. After the process is completed, the unreacted resin is discarded, whereas the post-curing treatment, such as heating, is applied to achieve the desired performance. This principle can apply to all SLA processes but differs due to the light's direction or the irradiation method [1]. Fig. 2 shows the working principle of stereolithographic printing [32].

2.1. History

SLA emerged in the 1970s when Swainson introduced a patent that utilized two intersecting radiation beams to fabricate 3D objects [38, 39]. Following this, Herbert, in 1982, introduced a system that construct objects layer-by-layer using photo-curable materials [40]. However, it is agreed that the starting point of modern stereolithography was the work of C.W. Hull in 1986 [14]. According to his approach, 3D objects were constructed layer by layer with photo-sensitive materials cured by ultraviolet (UV) radiation [14]. Huang et al. [32] have categorized these reported approaches into four generations depending upon their development. The first generation is based on the Charles Hull approach, which uses photo-sensitive polymers to build 3D objects cured by a light source [14]. To overcome the low efficiency of the Hull approach, projection stereolithography was introduced in the second generation, which cures each layer using photo mask technology [41,42]. The third generation emerged in 2015 when Tumbleston [43] developed a

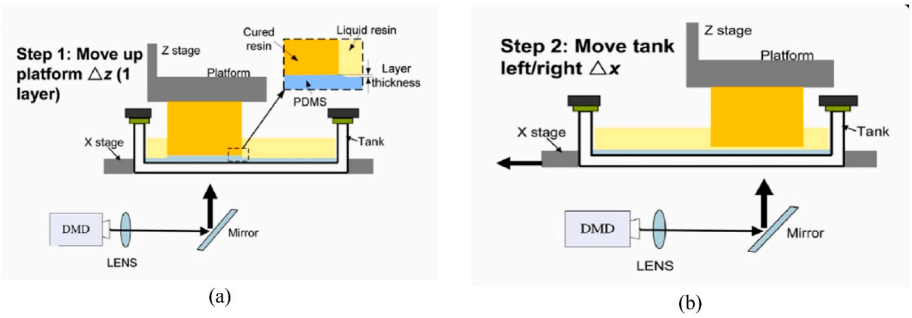


Fig. 5. Demonstratsmask projection stereolithography a two-way movement process (a) The platform movement is upwards along the *Z-axis*- have one layer of thickness, (b) the tank moves horizontally on the X-axis (Adapted from Ref. [7] Coprights © 2021 Elsevier B.V. All rights reserved.).

continuous liquid interface process (CLIP), which surpassed the other approaches in terms of speed. The fourth generation was volumetric stereolithography, which produces 3D objects with complex aperiodic 3D volumes in seconds [32]. The four generations are explained in Fig. 3.

2.2. SLA processes

Modern stereolithography, introduced by Charles Hull, has undergone four generations of innovations to enhance stereolithography capabilities [32]. This leads to the production of intricate systems that widely differ in scale and material composition [1,20]. This section reviews all the processes described above in each generation.

2.2.1. Laser scanning stereolithography

Laser scanning stereolithography involves employing a technique where a focused laser is directed over a resin surface to solidify the irradiated resin into a 3D structure [46,47]. Several researchers used galvanometric mirrors to handle the laser beam, but it led to the defocusing of the light beam and optical errors [46,48]. So, an alternative approach, the constrained surface technique, was employed [49,50]. In this setup, the laser point remains stationary, whereas the X-Y transition stage shifts the platform or the entire optical system when the object is being printed [51]. A glass window is used through which the light beam focuses on the resin to control the layer thickness. The advantage of the glass window is that it prevents the spread of resin to the polymerized part, but the drawback is that the resin then adheres to the window [52,53]. Miwa et al. [54] introduced a free surfacing scanning stereolithography to address this issue. This process is highly effective in fabricating microstructures as the resin has no adhesion issue. Another distinctive photopolymerization method is the two-photon polymerization (2 PP) [52,54,55] process, also known as direct laser writing (DIW), which utilizes near infra-red femtosecond laser pulses focused on a minute spot, causing polymerization through the simultaneous absorption of two photons [53]. This technique fabricates micro/nano-scale structures [56] with sub-100-nm resolution (structures smaller than 100 nm) [57] and surface roughness below 10 nm [58]. The process of 2 PP involves precise laser focusing, surpassing the diffraction limit, thus making 2 PP a highly versatile tool for designing and manufacturing complex 3D structures like microactuators and microswimmers [59,60]. The three-dimensional shape and structural heterogeneity of a microactuator have a significant impact on its performance. 2 PP provides exceptional flexibility in customizing the features, making it a promising technique for creating novel stimulus-responsive devices [61,62]. Nonetheless, two-photon polymerization faces the challenge of time consumption as they fabricate structures layer by layer [63]. The processes of the first generations are shown in Fig. 4.

2.2.2. Projection stereolithography

Projection stereolithography allows continuous layer printing through a single exposure by projecting mask patterns onto the resin surface. The concept of using photomasks was proposed in the 1980s and 1990s by Fudim [42] and Pomerantz [41], respectively. The process involved several steps: depositing the resin, illuminating it with a mask of a single cross-section, removing the uncured resin, filling and curing the resin, then grinding the surface to achieve a smooth layer and repeating the process till the object is completed [42]. However, this process required many masks and precise mask alignment, making it time-consuming. Dynamic masks were introduced to overcome these drawbacks. Bertsch et al. [64] used dynamic masks or liquid crystal displays (LCD) to set each pixel to a transparent or opaque state by changing the molecule's orientation. Although dynamic masks were superior to the previous processes, their inherent shortcomings included low UV light transmission, poor contrast, and large pixel size [65].

Another projection process that emerged involved the use of a digital micromirror device (DMD) as a dynamic mask [66,67]. DMDs are an array of several mirrors that are controlled independently to generate specific mask patterns. Initially, Bertsch et al. [68,69] reported using DMD in stereolithography while utilizing a light source emitting visible wavelengths. Subsequently, C.Sun et al. [66] introduced a DMD-based stereolithography system, which utilizes a UV light source and UV-curable resin to fabricate intricate 3D microstructures with a minimum feature size of 0.6 μm [32]. Similarly, Pan et al. [70] utilized a digital mirror device for mask-image-projection-based stereolithography (MIP-SL), offering a faster alternative to existing methods for selectively curing liquid photopolymer resin by managing the projected light. They explored a bottom-up projection system in MIP-SL, employing a two-way linear motion approach, which results in uniform thin layers. The finding demonstrated that utilizing a bottom-up approach for MIP-SL is an efficient and effective method that manufactures moderate-sized particles in minutes. Fig. 5 illustrates the mask image projection stereolithography based on a two-way approach as part a and demonstrates the process.

2.2.3. Continuous stereolithography

While projection stereolithography involves simultaneously curing each layer, it is not a continuous manufacturing process. Hence, it has drawbacks, including low printing speed and time consumption. Continuous liquid interface production (CLIP) was developed to overcome these drawbacks [71]. CLIP develops an oxygen-containing dead zone that inhibits photopolymerization between the window and the polymerization part, allowing that part to be continuously exposed [43,71]. The dead zone is crucial for continuous printing, ensuring a constant layer of fresh resin beneath the printed part. The flow of oxygen can regulate the thickness of the dead zone. Additionally, the motion of the build platform is continuous, moving slowly to match the resin curing rate [21,22]. This eliminates the final two steps of traditional stereolithography, enabling parts to be drawn out of resin continuously. This enhances the surface quality and significantly boosts printing speeds, reaching up to 500 mm/h [43]. A thermal-controlled continuous

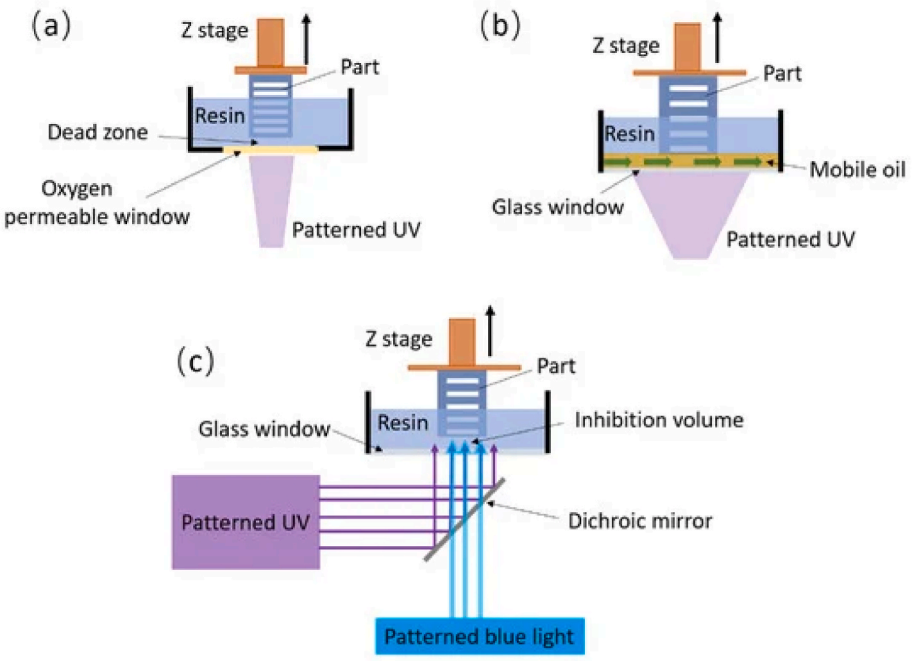


Fig. 6. The Flowchart for continuous stereolithography processes. (a) The continuous liquid interface production (CLIP) process schematic diagram. (b) The high-area rapid printing (HARP) system schematic diagram. (c) Dual-wavelength Continuous stereolithography schematic diagram (Adapted from Ref. [32] Coprights © 2020 by the authors. Licensee MDPI, Basel, Switzerland).

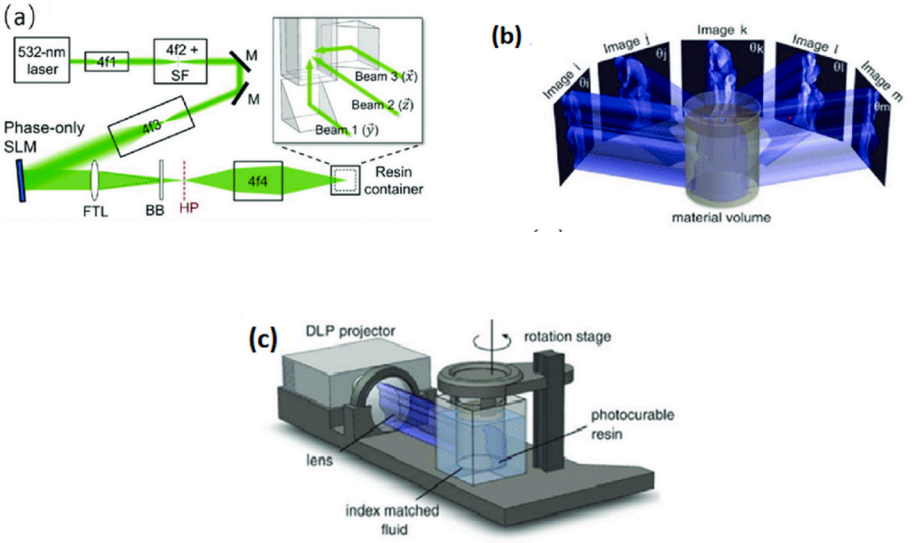


Fig. 7. (a) Volumetric stereolithography, (b) Tomographic volumetric stereolithography concept and (c) The tomographic volumetric stereolithography system (Adapted from Ref. [32] Coprights © 2020 by the authors. Licensee MDPI, Basel, Switzerland).

approach was introduced by Chad A. Mirkin [72], which reduced adhesive forces between the window and curing objects by utilizing a mobile liquid interface. The interface minimizes interfacial adhesion and dissipates heat, allowing high-end rapid printing without thermal limitations. Similarly, Scott et al. [73] demonstrated a continuous stereolithography system that employs dual-wavelengths. One wavelength of light hinders polymerization, while the second wavelength triggers resin polymerization. An inhibition region is formed between the window and curing part intensity by adjusting the light source, effectively eradicating adhesion and facilitating continuous printing. These innovations represent an advancement in stereolithography with rapid printing speed and reduced adhesion limitations. The continuous stereolithography processes are presented in Fig. 6.

2.2.4. *Volumetric stereolithography*

Volumetric stereolithography builds 3D structures in one operation by forming 3D volumes [74]. Shusteff [45,75] designed a volumetric stereolithography system using three orthogonal beams projected into a photo-sensitive resin, creating 3D geometries in a single exposure. Timothy F. Scott's [76] team extended the concept of dual-wavelength by utilizing two perpendicular irradiation patterns at blue and near-UV wavelengths. Each wavelength independently influences polymerization initiation or inhibition to achieve volumetric 3D printing. Tomographic volumetric stereolithography, a process that was

Table 1
Comparison of the four generations of stereolithography.

Stereolithography	Printing Speed	System Resolution	Printable Size	Light Source	Suitable Material	Techniques	Advantages	Challenges	Commercial Printers	Ref.
Scanning	Scanning speed (mm/second): ranges from hundreds to thousand	Resolution (microns): A few microns (~100 nm for 2 PP system)	From tens to hundreds of millimetres	UV light (Femto second Infra Red laser source for 2 PP)	Acrylate, methacrylate, epoxy, vinyl monomers	Constrained surface technique Free surface technique Two-photon polymerization (2 PP)	Fabricating micro/nano-scale structures High Resolution	Adhesion of resin, Time-Consuming Low building speed	Formlabs Form 3, 3D Systems SLA 3D Printers	[47, 53–55]
Projection	Speed (mm/hour): From Tens	(microns): A few (mainly more than 5)	Tens of millimetres (DMD-based can print up to 0.6 μm)	UV/Visible light	Acrylate, methacrylate, epoxy, vinyl monomers	Projecting mask Dynamic mask Liquid crystal display (LCD) Digital micromirror device (DMD)	High efficiency	Single-use masks, Precise mask alignment, Time-consuming, Pixel size large, Transmission in Ultra Violet very poor, Contrast Poor	LumiForge, Phenom MSLA Printer	[64–66, 83,84]
Continuous	Speed (mm/hour):Hundreds	(microns): A few (mainly more than 5)	Tens of millimetres	UV/Visible light-Dual Wavelength	Acrylate, methacrylate, vinyl monomers, epoxies	Continuous Liquid Interface Production Thermal controlled continuous stereolithography	Continuous process, Rapid printing speed	Minimize interfacial adhesion Low-viscosity resin required	Carbon 3D Printers	[43,71, 73]
Volumetric	Greater than 10^5 mm per hour	(microns): 80–300	Tens of millimetres	UV light- Dual wavelength	Acrylate, methacrylate, vinyl monomers, epoxies		Direct 3D printing Smooth surface Fast printing	Time-consuming, Continuous Refilling of the tank		[75,76, 78–81]

introduced by Hayden K. Taylor et al. [77] and Loterie et al. [78]. This approach leverages tomographic reconstruction to create complex, volumetric parts. This technique simultaneously illuminates an entire volume of photosensitive resin within a cylindrical container [79]. By projecting distinct images of the object captured from precise angles while continuously rotating the cylinder, the resin gets exposed from all sides and angles, resulting in a 3D light dose that causes the resin to become gelatinous, ultimately fabricating desired objects as a whole volume [11,45,80]. Volumetric Stereolithography directly constructs the object in 3D, in contrast to the other AM approaches constructing the 3D structure through successive 2D layers [81]. This technique has the potential to manufacture complex and intricate parts while achieving exceptionally high printing speeds [79], capable of producing structures up to 55 mm in just 30–300 s, making it ideal for efficient, high-volume production [82]. However, the resolution of the printed parts is currently limited to a range of 80–300 μm, depending on various physicochemical factors such as the viscosity and reactivity of the resin, as well as the distribution and exposure of light within the resin [78]. Nevertheless, it is expected to evolve further in terms of resolution, material, manufacturing process and physical system. The processes are presented in Fig. 7. Table 1 shows the comparison between the four generations.

2.3. Other processes

Researchers have introduced various other processes and systems throughout the evolution of stereolithography processes. One such innovation is multi-material stereolithography, based on the construction of 3D objects with multiple materials [85–87]. Multi-material stereolithography enables the integration of multiple materials (typically two or three) into a unified 3D printed structure by switching between materials during printing. Choi et al. [88] outlined the designs and implementations of the MMSL machine, introducing innovative materials and methodologies. The MMSL machine required the construction and development of various components, including a new frame, platform assembly, a rotating Vat carousel system, an automatic levelling system, and a custom control system for automated process control. The successful fabrication of several multi-materials created distinctive, functional, and visually appealing objects. While the system could produce complex multi-material parts, it has some limitations, including restriction to low-viscosity resins due to its deep-dip coating approach; hence, this prolongs fabrication duration and introduces deformations in the structure due to cumulative registration errors [89].

Another innovative approach was developed in the Centre for Rapid and Sustainable Product Development of the Polytechnic Institute of Leiria (Portugal) by researchers called Stereo-Thermal-Lithography (STLG) to produce functionally graded components compromised of multi-material [90]. The method utilized ultraviolet (UV) radiation and thermal energy to trigger a polymerization reaction in a resin containing both photo- and thermal initiators [91]. The reaction only commences with a specific combination of UV radiation and thermal energy, so the concentration of the initiators is carefully measured. Thermal stereolithography consists of four subsystems.

- **Subsystem A** - uses UltraViolet (UV) radiation to solidify a photoinitiator-containing resin, the same as the conventional stereolithography.
- **Subsystem B** - uses InfraRed (IR) radiation to solidify a thermal initiator resin.
- **Subsystem C** - then utilizes both IR and UV radiation to cure a resin containing both initiators.
- **Subsystem D** - incorporates both IR and UV radiation to cure resin containing both initiators [36].

This approach allows the generation of radicals, deeper penetration of radiation into the polymer, and increased reaction rates, leading to

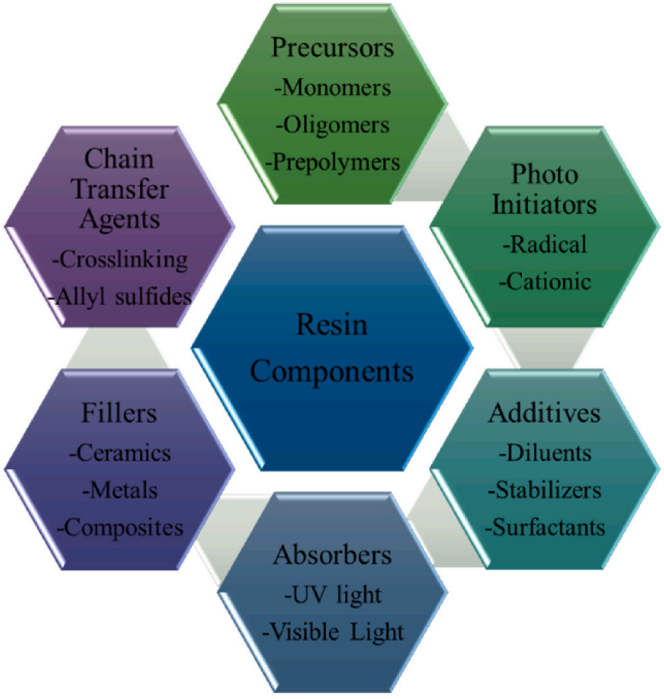


Fig. 8. Resin Components For Stereolithography [21].

improved accuracy and tunability of the produced models. Additionally, thermal stereolithography includes a rotating multi-vat mechanism, facilitating the fabrication of multi-material structures, including engineering prototypes, tissue engineering scaffolds, biomedical components, and micro-functional metallic or ceramic parts [90].

Colour stereolithography aims to fabricate resin prototypes with multicolour areas. This process generates semi-transparent prototypes that reveal the internal structure based on the colour and shape of coloured areas [92]. Colour stereolithography has been adopted to add vibrancy and visual appeal to prototypes and enhance their aesthetics. Additive manufacturing processes are not inherently colour-repetitive due to challenges like incompatible dyes, pigments with resins, intricate setups, and cost considerations [93]. Swaelens and Vancraen [94] worked on highlighting parts of the structure during the building process. They used additives containing clear resin that changed colour on exposure to high doses of UV radiation. After curing each layer, the areas that needed highlights were rescanned with a laser at a slower speed to deliver high doses of UV radiation, resulting in highlighted features. Gong and Wang [92] invented a colour stereolithography method which was based on the time-pressure dispensing process. They had designed a positioning-controlled system to control the motion of the syringe extruding colour resin. Some possible applications of colour stereolithography include concept modelling, texturing, footwear and clothing design, packaging, etc. [93].

2.4. 4D printing for SLA

Complex 3D functional tissues can be printed using a 3D bioprinter, together with biocompatible materials and even living cells and are employed in a variety of biomedical applications, including drug screening and delivery, wound healing systems, transplantation, bio-actuators, bio-robotics, and bio-sensors [95,96]. However, 3D printing has a huge drawback in the construction of immobile structures that are unable to adapt to their surroundings or change shape [97]. To overcome this limitation in 3D printing, 4D technology has been developed that extends the capabilities of traditional 3D printing by introducing a fourth dimension, i.e. time. This enables bio-printed constructs to evolve and adapt over time, responding to both internal and external stimuli like temperature, pH, light, moisture, electrical and magnetic fields, etc [98–102]. 4D printing uses materials that are programmed to alter their properties in a controlled manner when exposed to external stimuli. Common 4D printing techniques include extrusion-based methods and vat polymerization, such as stereolithography (SLA) and two-photon polymerization (2 PP) [97,103]. SLA is particularly well-suited for 4D printing due to its ability to produce high-resolution and high-quality surface finishes [104]. Whereas 2 PP is also employed to construct microswimmers, microactuators and Volumetric stereolithography to create 4D structures in seconds [60,105]. 4D printing, being a relatively new technique, constructs structures that adapt to their environment and respond to stimuli and offers promising solutions for creating more functional and personalized medical devices but it faces several challenges, including material development, design constraints and manufacturing efficiency [98].

3. SLA material components

The resin has essential components for stereolithography to achieve high printing quality, as shown in Fig. 8.

- **Precursors** - Monomers/Oligomers or prepolymers form a network upon polymerization.
- **Photo initiators (PI)** -to initiate the polymerization reaction.
- **Additives** - diluents, stabilizers, or surfactants.
- **Absorbers** - to control light penetration.
- **Fillers** - fabrication of ceramics or metals powder to resin
- **Chain Transfer Agent** - an agent that modifies the cross-linking extent.

However, SLA resin formulation is versatile and adaptable to meet the specific requirements and desired properties that are sought [21,22,106].

3.1. Precursors

Monomers and oligomers or prepolymers are the liquid molecules in the resin that undergo a polymerization reaction upon exposure to light to form a solid 3D structure. The type of precursor used depends upon the intended properties and applications [106,107]. SLA formulations are mainly based on poly-acrylates or their derivates due to their heightened reactivity, low cytotoxicity, and high heat resistance [107]. By adjusting the number of reactive groups or incorporating diverse oligomers, the acrylate-based or meta-acrylate resins can be customized to achieve desired mechanical and thermal properties [108,109]. However, acrylate resins have a potential drawback due to their tendency to shrink during printing. To mitigate this issue, acrylates are mixed with methacrylate. Another issue with acrylates is oxygen sensitivity, which can hinder the polymerization reaction.

In contrast, epoxy resins follow a cationic polymerization reaction; hence, they have a different curing mechanism. While they require longer reaction time and can be inhibited by moisture, these resins have significantly lower shrinkage and offer stability against oxygen. Multi-material resins have been introduced to capitalize on the benefits of both resins. Combining both resins results in materials with fast curing and low shrinkage properties [110–112]. Several other polymers that follow cationic polymerization include propenyl ethers, vinyl ethers, cyclic acetals, and furfural and siloxanes [113]. Moreover, the monomer and oligomer molecular weight influences the viscosity of the polymer. High molecular weight causes an increase in the viscosity of the resin, which results in longer setting times. In this case, diluents are added to adjust the viscosity [114,115].

3.2. Photo initiators (PI)

Photoinitiators are crucial components of the resins because some

prepolymers cannot be activated upon irradiation [116]. A low molecular weight organic photoinitiator is added to the resin to generate reactive species under laser exposure, thus initiating the curing reaction [22]. When PI are exposed to the appropriate wavelength, they get excited and initiate a polymerization reaction by targeting the functional group in the monomers/oligomers. Once the functional group breaks, each carbon can react with other carbon atoms, forming a different monomer species [108,117,118]. The type of PI used depends upon the nature of the precursor. When exposed to light, free radical photoinitiators form free radicals that react with acrylates and methacrylates [119]. Meanwhile, when exposed to light, the cationic photoinitiators produce acids and react with vinyl ethers and epoxides [113]. The selected photoinitiators have significant implications on the reaction kinetics, required light exposure, conversion rates, cross-linking density, and mechanical properties [108,117,118,120]. Therefore, an ideal photoinitiator should exhibit high initiation efficiency, good solubility in the prepolymer, stability during storage, absence of polymer degradation, and non-toxicity [36].

3.3. Additives

Additives are added to ensure the prolonged shelf life and stability of the resin [106]. An increase in the proportion of solid volume increases the viscosity of the resin, impacting the flow behaviour of the resin [121]. These modifications influence the coating mechanisms and the mechanical force required for the elevation of the building platform. Various approaches can be employed to mitigate the issue, including using diluents, temperature adjustment or thinning behaviour [122–124]. Diluents are primarily added to modify the viscosity of the resin to maintain optimal wetting and impregnation of the resin. The diluents are selected based on their reaction with the prepolymer and integrate into the solidified resin but are unreactive while in storage [107].

Similarly, rheological additives and stabilizers are added to reins to enhance solid loading and ensure the extended shelf-life of slurries when printing for a longer period. Agglomeration and sedimentation of particles are necessary for continuous and homogenous printing of ceramics and metals, maintained by adding dispersants or surfactants like oleic acid or phosphine oxides [125]. A study conducted by Kehang Li et al. [126] examined oleic acid (OA), stearic acid (SA) and Poly(Acrylate Ammonium) (PAA-NH4) as surfactants for formulating suspensions of HDDA (hexanediol diacrylate). It was concluded that the surfactants enhanced the stability of the HDDA suspension and decreased its viscosity. Among the three surfactants, oleic acid (OA) was the most effective as a fatty acid; it exhibited superior affinity in the hydrophobic medium of the suspension.

3.4. Absorbers

Absorbers are added to the resin to control the light penetrating the resin and the curing depth, which is necessary for the printing of complex objects with undercuts, as precision in curing depth is crucial to prevent excessive curing and loss of specific features [106,127,128].

3.5. Fillers

To produce composite materials, reinforcing metals or ceramics are added to the resin in powdered form, followed by printing and subsequently debinding and sintering. Debinding is performed to remove organic resin through pyrolysis, while the sintering process involves thermal treatment to form dense structures. High filler content minimizes shrinkage, leading to accurate geometrically printed structures [38,129,130]. In a highly dense matrix where the particle size ranges to the wavelength of the light used, scattering is the primary interaction mechanism to determine curing depth and affect resolution. A common strategy to reduce the scattering effect is to decrease the refractive index between the filler and the matrix [131]. Fillers in resin can also be in the form of composites, where the organic part is retained in the final product. Integrating SLA resin with nanoparticles enhances the mechanical, thermal, optical, and even electrical properties of the polymers [132,133].

3.6. Chain transfer agents

Chain transfer agents are necessary for SLA resin to regulate network formulation by modifying the extent of cross-linking, which in turn enhances the properties of the printed structure. Addition-fragmentation chain transfer (AFCT) agents, such as Allyl sulfides, are used to control the molecular weight in free radical polymerization processes [134,135]. According to Kloxin et al. [136], when allyl sulfide AFCT is integrated with thiol-ene polymerization, a chain transfer process occurs in the presence of free radicals and hydrogen is abstracted. They used allyl sulfides and methacrylate or vinyl monomers to form a cross-linking network with higher cross-linking density and reduced shrinking stress. Another study by Gorsche et al. [137] outlined a process to incorporate β-allyl sulfone as an AFCT agent to enhance the toughness/strength of cross-linking networks made up of methacrylate. This is achieved as β-allyl sulfone attaches to free radical species directly, leading to the sulfonyl radical's fragmentation, thereby continuing the chain transfer process.

Recently, Dall et al. [138] investigated the importance of coupling different chain transfer agents CTA with BADGE resins at high temperatures and how it enhances their reactivity. Trimethylolpropane (TMP), polypropylenglycole (PPG), and polytetrahydrofurane (PTHF) were selected as CTA and infused with the resins. The chain transfer agents led to a linear decline in the glass transition temperature as the content in the photo-curable formulation increased. This trend was due to the hydroxyl (OH) groups in the anionic mechanism, which caused an increase in the polymer network's mobility but decreased the cross-linking density. Subsequently, Dall et al. could print 3D structures with high thermomechanical properties at 80 °C. Chain transfer agents in resins are now being studied to develop low-shrinkage composites for medical purposes. Grob et al. [139] assessed the potential of ethyl-2-(tosylmethyl)acrylate (ASEE) as a chain transfer agent for the development of dental composites with low-shrinkage. They prepared three composites with ASEE concentrations of 10, 20, and 30 mol% in the monomer matrix [containing 1, 6-bis-[2-methacryloyloxyethoxycarbonylamino]− 2,4,4-trimethylhexane (UDMA, mixture of isomers), 1,10-decanediol dimethacrylate (D3MA)] and various photoinitiator systems. Investigating key factors, including flexural strength and modulus, Glass transition temperature Tg, FWHM, cross-link density and shrinkage force, revealed the effect of photoinitiators on the composites. The ASSE composite with photoinitiator CQ/EDAB/SC-938 (0.33 wt%/0.60 wt%/1.00 wt%) formulation proved to be suitable for dental application due to its negligible shrinkage force, high double bond conversion (DBC), suitable working time and enhanced mechanical properties without any effect on the concentration of the monomer in the sample. To learn more about the effect of photoinitiators on monomers, refer to Ref. [139].

3.7. Material requirement for SLA

The manufacturing process of SLA involves preparing a photoreactive medium by combining a photocurable resin with suitable reinforcements before exposing it to the laser. Upon polymerization, the photopolymer organizes around the reinforcements, forming a solid matrix [140]. For the resin to solidify effectively and be suitable for various applications, the suspension must meet specific criteria, including particle dispersion, resin viscosity, photosensitivity, and resolution. The particles must be stably dispersed within the resin, while the viscosity should remain below 5 Pa.s to ensure efficient layer recoating and minimize the time required to create a fresh, even layer. Additionally, the UV-curable suspension must have adequate curing

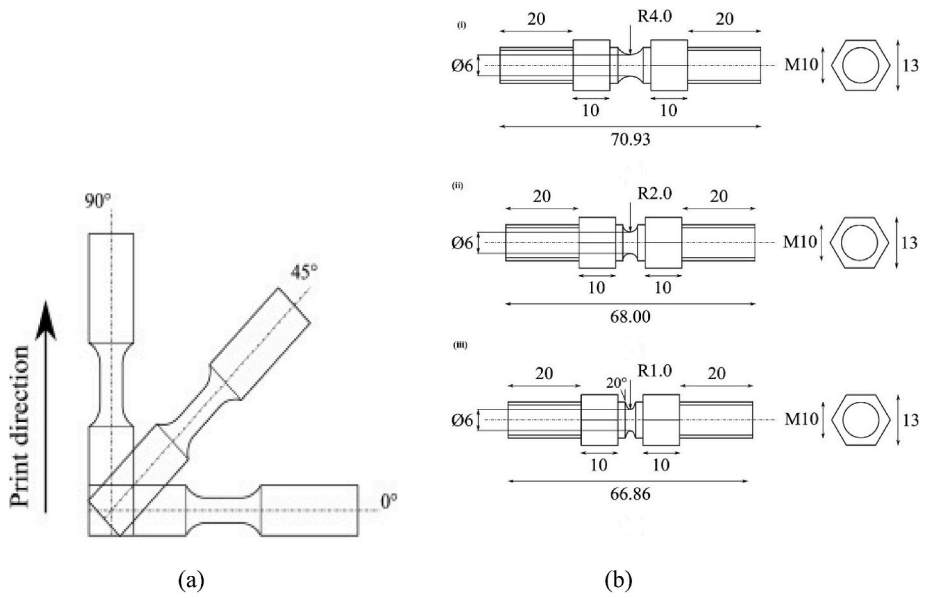


Fig. 9. (a) Demonstrates the different print orientations for anisotropy tests (b) demonstrates the specimens for notched compression with notch radii (i) 4.0 mm, (ii) 2.0 mm and (iii) 1.0 mm (Adapted from Ref. [144] Copyrights © 2023 The Authors. Published by Elsevier Ltd).

depth and resolution. Curing depth is essential to avoid prolonged fabrication times and results in smooth surfaces [141]. The addition of particles significantly alters the resin's photopolymerization behaviour due to light scattering and absorption. It's also imperative to minimize shrinkage during the liquid-to-solid phase transition.

Traditionally used polymers include epoxy, acrylate resins, and thermoplastic elastomers. Acrylate resins offer several benefits, such as low viscosity for smooth spreading, high photosensitivity, resistance to oxygen inhibition, low critical energy requirements, adjustable mechanical properties, and relative insensitivity to temperature and humidity changes. However, their use has been somewhat constrained by issues like poor dimensional stability and significant volume shrinkage, with distortion caused by post-curing shrinkage being a major concern. On the other hand, epoxies have been developed with superior characteristics, including exceptionally low volume shrinkage and improved dimensional stability. Whereas epoxies have been developed to offer superior properties [13,141].

4. SLA applications

Additional information on the processing of photo-curable material using stereolithography (SLA) technology will be provided in the following sections.

4.1. SLA for mechanical applications

The stereolithography technique has recently been utilized to process polymers and polymer composites for mechanical applications. The selection of process parameters and post-curing treatment plays a vital role in influencing the quality of the final product. Several researchers have investigated how these process parameters impact the mechanical properties of 3D-printed parts. Liu et al. [142] examined how the mechanical properties of photo-curable epoxy resin (NH-90) specimens are affected by variations in build orientation printed through stereolithography (SLA). The samples were 3D-printed in various orientations and were analyzed for compression, impact strength and tensile properties. They discovered that the specimen printed on the edge exhibits higher ultimate tensile strength and lower failure strain than the flat-printed specimen. The maximum ultimate tensile strength (UTS) of 37.2 MPa was observed for the specimen printed on the edge, while for the specimen printed flat X axis at 900, Young's modulus of 908 MPa was observed. In comparison, the impact strength conducted on the X and Y axes of both printed specimens revealed that orientation does not significantly affect impact strength.

Casafont et al. [143] investigated the impact of anisotropy on the compression behaviour of acrylate resin using 3D objects printed via mask image projection SLA. The study employed various techniques to gather stress-strain data and produced six orientation specimens without curing treatment. The study concluded that the maximum anisotropic effect was observed using an extensometer, showing a 32 % deviation (PD) and a 39 % coefficient of variance (CV). Ruben Løland Sælen et al. [144] presented the characterization of commercial SLA resin. They investigated the effect of triaxiality, strain rate and print orientations through tensile strength and compression tests. It was concluded that (a) the material exhibits a typical glass polymer behaviour, initially being elastic and then transitioning into inelastic flow and hardening due to polymer chain orientation, (b) the hyper-viscoelastic constitutive model accounted for the major mechanical properties; however, some deviation was observed for certain specimen geometries, and (c) the constitutive model demonstrated the trends from the compression test with simulated force-displacement curves matching the experimental data however, the dynamic tests had underestimated elastic stiffness as the model had overestimated the force levels and did not consider the linear viscoelastic effects. Fig. 9. (a) Demonstrates the print orientations for anisotropy tests, and (b) demonstrates the specimens for notched compression with notch radii (a) 4.0 mm, (b) 2.0 mm and (c) 1.0 mm [144].

Introducing additives has proven effective in enhancing photosensitive material's curing process and mechanical properties. In their work, Alamdari et al. [145] employed a surface energy-reducing agent (SERA) in the liquid bridge micro-stereolithography technique. They assessed the changes in adhesion force and mechanical properties by varying the concentration of the monomers, light absorbers, photoinitiations, and SERA in the polymer resin. However, the introduction of SERA led to a reduction in the elastic modulus. Zhang et al. [146] described the fabrication of lignin-based methacrylate resin with enhanced mechanical properties. Lignin was incorporated into methacrylate resin in different concentrations, and the mechanical properties of the non-post-cured and post-cured printed objects were determined. Incorporating lignin significantly improved the tensile strength-increasing it by 46–64 %, and Young's modulus increased by 13–37 % for post-cured specimens. The tensile strength for the 0.4 %

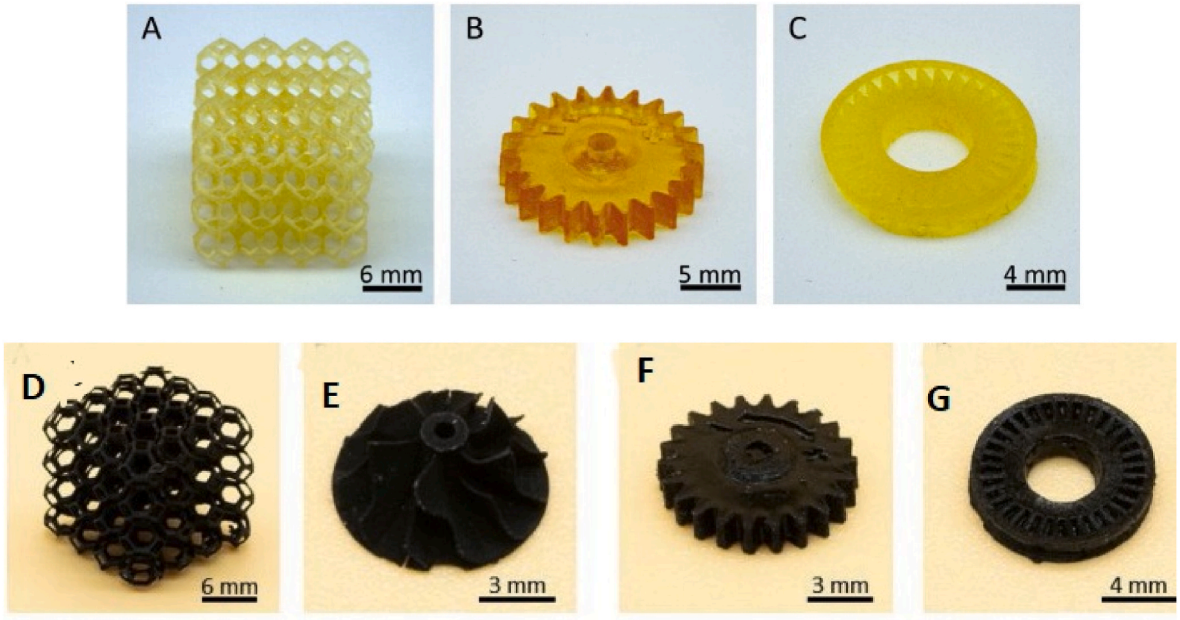


Fig. 10. Display Structures of Thermosets produced from 3 distinct polymer classes: (A) polysiloxane SPR212 printed Structures, (B) Represents a polycarbosilane SMP10 fabricated wheel structure, (C) Polycarbosilazane Durazane 1800 printed turbine structure, (D, E) SiOC used to fabricate a cell structure and a turbine impeller and (F, G) A turbine and cog wheel composed of SiCn(O) (Adapted from Ref. [150] Copyrights © 2019 Elsevier B.V. All rights reserved.)

kraft lignin incorporated sample was 49.0 MPa, about 60 % higher than the control sample. This study also revealed that the composites exhibited a rougher fracture surface, contributing to stress dissipation and enhancing mechanical properties.

Post-processing stages have a considerable impact on the mechanical properties of printed objects. Nowacki et al. [147] investigated the effects of post-processing on the mechanical properties of an element produced by mask stereolithography (mSLA). Basic translucent green UV-sensitive resin was fabricated and subjected to post-processing steps of washing with 2-propanol and post-curing with UV light at varying durations. Following post-processing, the samples were projected to acclimate for three days before they were analyzed for static tensile, static bending tests, and profile-graphometric or hardness analysis. It was observed that longer post-curing significantly improves the sample's tensile and flexural strength and increases the specimen's hardness. The effect of washing was ambiguous, as a decrease in strength was observed for prolonged washing time. Ahmad et al. [148] conducted a similar study on a photopolymer (PR 48). The research aimed to comprehend how print orientation, layer thickness, UV post-curing time, and post-curing temperature influence printed samples' elastic modulus and hardness. The investigation included assessing the resolutions of 50 μm and 100 μm at 0°, 45°, and 90° orientations. The 100 μm exhibited significant anisotropy in elastic modulus and hardness at 0° and 90° in contrast to the 50 μm sample. This highlighted the consideration in choosing optimal print parameters for desired outcomes, as printing parameters significantly impact the mechanical properties of the 3D printed structures.

Despite being able to print intricate structures with limited production times and reduced material wastage, printing nanofiller-reinforced lattices by SLA is challenging. Markandan et al. [149] studieed stereolithography-printed graphene-polymer octet-truss lattices to investigate the various factors impacting the mechanical properties of the graphene-polymer composite, including filler concentration, solvent addition, and post-printing baking temperature and duration. Enhanced mechanical properties of the composite were observed with low graphene concentrations due to the alignment of graphene platelets into the polymer matrix during printing. The graphene polymer composite fabricated exhibited anisotropic mechanical properties and increased the material's modulus and strength with post-baking. This composite was proposed for applications that demand a high level of stiffness and strength.

Wang et al. [150] pioneered the application of stereolithography (SLA) in the additive manufacturing (AM) of ceramics to bolster the chemical and mechanical characteristics of resultant polymers. Three preceramic polymers were selected — polysiloxane SPR212, polycarbosilane SMP10, and polysilazane Durazane1800 to fabricate polymer-ceramic compositions of SiOC, SiC, and SiCN, respectively. The SiOC ceramic exhibited a mean reduced modulus of 106 ± 6 and hardness of 12 ± 1 GPa. The compressive strength of 216 MPa was recorded with a theoretical density of 97 %, characterized by minimal porosity and defects. Similar to the work of Wang, Chauhan et al., 2022 [151] explored the potential of newly developed Poly(ethylene glycol) diacrylate (PEGDA)/bioceramic composites fabricated through stereolithography for load-bearing application in tissue engineering. PEGDA resins were blended with 1%wt of hydroxyapatite (HAP), calcium phosphate (CaP), and graphite. The resulting printed structures underwent characterization and assessment for mechanical, tribological, physical, and biological properties. The results revealed PEGDA/bioceramic composites suitable for 3D printing by SLA and as biocompatible, non-toxic, and conducive to cell growth. PEGDA/Gr sample had the highest compressive strength and lower wear and frictional force. Meanwhile, PEGDA/HAP exhibited the highest average tensile strength. Overall, the composite had improved properties, making it suitable for tissue engineering applications such as cartilage and bone repair. Fig. 10 represents the parts fabricated through SLA. Parts (A), (B), and (C) show structures printed with polysiloxane SPR212, polycarbosilane SMP10 and polycarbosilazane Durazane1800, respectively. Part (D, E) shows cell structures with SiOC, whereas part (F, G) shows SiCN(O) printed structures [151].

Multimaterial 3D printing is a relatively recent concept where a blend of multiple polymers is synthesized to create innovative material for various applications. Zainuddin et al. [152] studied the mechanical and structural properties of polyhydroxy butyrate (PHB) blended with polyurethane acrylate (PUA), formulating PHB/polyurethane acrylate (PUA) blend resins with varying concentrations of PHB at (0 wt%, 6 wt %, 12 wt% and 18 wt%). The tensile and impact strength of these

Table 2
SLA process for mechanical applications.

Materials	Additives and Reinforcements	Printing Parameters/Material Concentrations	Properties Improved	Ref
Epoxy-based photocurable resin NH-09	Nil	• Build orientation: flat X direction, flat Y direction, on edge X, on edge Y, flat X specimens (from 0° −90°), Flat XY direction, and on edge XY • Light wavelength - 355 nm • Laser Critical exposure- 10 mJ/cm2 • Penetration power of light - 110 μm	• Highest attained Youngs Modulus for specimen at X-axis flat at (90°) - 908 MPa • Impact Strength maximum was achieved on edgeX - 22.8KJ/m2 • Compressive strength highest attained at flatX (0°) - 25.9 MPa	[142]
Acrylate photoactive resin	Nil	• Six Printing directions along the axis: (X, Y, Z, XY, XZ and YZ) • Resolution: 62.5 μm • Layer thickness: 75 μm	• Percent deviation (PD) decreased up to 68 % • Coefficient of Variance (CV) decreased Upto 105 % • Elastic modulus ratios (Ei/Ej) range from 0.81 to 0.97	[143]
Photoreactive methacrylate resins (RS-F2-GPCL-04) methacrylate monomer, oligomer and initiator	• Softwood kraft lignin • DMSO (Dimethyl sulfoxide, ≥99.9 %) • 2-Chloro-4,4,5,5-tetramethyl-1,3,2 dioxaphospholane, • endo-N-hydroxy-5-norbornene-2,3-dicarboximide	• 6 different Lignin concentrations were incorporated into resin at 0.0, 0.2, 0.4, 0.5, 0.8, and 1.0 wt % • Post curing wavelength of 365 nm • UV intensity of the post-cure device 280 ± 16 μW/cm2	• Post-cured samples exhibit higher ultimate tensile strength regardless of lignin concentration • more than 0.5 wt % lignin showed low tensile strength for non-post-cured samples • Post-curing significantly increases tensile strength; lignin concentrations of 0.2 and 0.4 wt % tensile strength increase by 52 % and 64 %, respectively • Lignin concentration increases Young's modulus samples up to 0.4 wt % • At 0.2 wt %, Young modulus increases by 26 % in the post-cured sample	[146]
Preceramic polymers: • Liquid methylvinylhydrogen polysiloxane (polyramic SPR-212) • Allylhydrydopolycarbosilane (StarPCS™ SMP10)	• Photo initiator: (BAPOs) phenyl bis (2,4,6-tri- methyl benzoyl) phosphine oxide • Photo absorber: Sudan Orange G • Radical Scavenger: Hydroquinone	• Wavelength- 450 nm • Layer thickness- 100 μm • Curing time - 30 min • Temperature for pyrolysis - 1100 °C	• 106 ± 6 GPa mean reduced modulus • 216 MPa compressive strength • theoretical density 97 % • hardness 12 ± 1 GPa	[150]
PEGDA Hydrogel Resin	• (BAPO) phenylbis(2,4,6-trimethyl benzoyl)phosphine oxide • Hydroxyapatite (HAP) • Calcium Phosphate (CaP) • Graphite (Gr) •	• Composite formulation PEGDA- 1% wt of HAP, CaP, and Gr • Assessment of PEGDA/Biocreamic composites	• Highest average tensile for PEGDA/HAP (2.15 ± 0.198 mpa) • Highest compressive strength attained for PEGDA/Gr (8.71 ± 0.39mpa) • Flexural strength improved by 218 % for PEGDA/HAP, 106 % for PEGDA/cap, and 93.2 % for PEGDA/Gr. • Hydrophilicity property enhanced with fillers. • Composites showed thermal degradation upto ~550 °C and thermal stability ~187 °C	[151]
Commercial SLA Resin	Nil	• Characterization of commercial SLA resin based on triaxiality, strain rate and print orientations	• The material exhibited typical glass polymer behaviour • Size effect governed the fractures in the material	[144]
UV-sensitive resin, basic translucent green resin	Nil	• Exposure Time: 8 s • Layer Thickness: 0.05 mm • Lift Distance: 5 mm • Resin Vat Temperature: 21–25 °C	• Longer post-curing improves tensile and flexural strength • Increases the specimen's hardness • Post-curing up to 30 min improves surface quality • Optimal Washing is 10 min, and post-curing for a minimum of 30 min.	[147]
Acrylate based resin • iso bornyl acrylate (IBXA), •1,6-hexanediol diacrylate (HDDA) bisphenol-A-ethoxylate (4) diacrylate (BEDA) weight ratio 4.5:4.5:1 by wt%	• f 2,2-dimethoxy-2-phenylaceto-phenone (DMPA) as PI (1 wt%) • Tinuvin 327™ as light absorber (0.15 wt%) • BYK-UV 3500 as surface reducing agent (0–20 %)	• Optimum SERA concentration - 5 wt % • Laser Wavelength - 365 nm • Irradiance light ~9.6 mW/cm2 • Exposure Time - 6 s • Thickness of Layer - 10 μm	• Elastic Modulus maximum achieved was 540 MPa	[145]
Polyurethane Acrylate (PUA)	• Polyhydroxybutyrate (PHB) • PHB: 0, 6, 12, and 18 wt% (in blend)	• Sample dimensions are 63.5 mm in length, 9.53 mm in width, and 3.2 mm in height. • Assessment of tensile testing, impact strength, structural and thermal evaluation	• 18 wt% PHB was optimal content in PHB/PUA blend compositions with a viscosity of 2172 cP • Crystalline Index (CI) was highest for 18 wt% PHB/PUA blend upto (22.05 %)- making it more brittle	[152]

(*continued on next page*)

Table 2 (*continued*)

Materials	Additives and Reinforcements	Printing Parameters/Material Concentrations	Properties Improved	Ref
			• Tensile and impact strength affected by PHB, with a p-value <0.05. • 12 wt% PHB/PUA blend 3D printed	
Photopolymer Resin • EBECRYL 8210 (aliphatic urethane acrylate): 39.8 wt% • SR 494 (Ethoxylated pentaerythritol tetraacrylate): 39.8 wt%	•PhotoInitiatior: TPO (Diphenyl (2,4,6- trimethlbenzoyl)phosphine oxide): 0.4 wt% • Dilutent reactive: Genomer 1122 (Monofunctional urethane acrylate): 19.8 wt% • UV blocker: 2,2-(2,5-thiophenediyl)bis(5-tertbutylbenzoxazole): 0.16 wt%	• Samples dimensions: 20 × 20 × 3 mm. • Resolutions: 50 μm and 100 μm • Print Orientation 0°, 45° and 90° • UV curing room temperature and 80 °C	• 100 μm resolution exhibit anisotropy in elastic modulus and hardness with highest value at 0° • Better Elastic modulus and hardness properties in in 0° and 45° orientation as layer thickness was high around 100 μm • For Lower layer thickness (50 μm) are better in 90° orientation • UV curing at 80 °C has improvement in mechanical properties	[148]
Grey resin • Methacrylate oligomers and monomers	•Graphene platelets size of 15–25 μm • Thickness of 6–8 nm	• Graphene loading 0.01 wt% to 0.1 wt % • Graphene-polymer composites, 5 mm × 5 mm × 5 mm cubes • Resolution of 0.025 mm • Octet truss lattices with 4 × 4 × 4 unit cells • Solid fraction of 0.31	• Porosity for the composites is8% as a whole • Graphene-polymer composites has anisotropic behaviour • Dispersing graphene in acetone decreases Young's Modulus and yield stress • An increase in curing increases Young's modulus and yield strength • Baking at 120 °C affects mechanical properties • 0.1 wt% graphene reinforced lattices had deteriorated mechanical properties	[149]
4-acryloylmorpholine (ACMO), acrylic acid (AA), poly (ethylene glycol) diacrylate (PEGDA)	•Photoinitatiaor: (TPO) Diphenylphosphine oxide •Ion (III) chloride, aluminum sulfate octadecahydrate •Dyes: Sudan III (red dye), Orange II (Yellow dye) •(EDTA) ethylenediamine-tetra acetic acid	• TPO 1.5 wt% and Sudan III dye with 0.025 wt% optimum • Curation parameters: UV chamber - 66 mW/cm^2 • Wavelength- 265–700 nm	• Post-treatment through simultaneous hydration and ionic coordination transform stiff drygel into soft hydrogel • adjusting chemical composition and ionic sources the hydrogels exhibits high level of mechanical tunability and self-weldability.	[153]
Monomer (ethoxylated trimethylolpropane triacrylate) Oligomer (difunctional aliphatic urethane methacrylate)	•Diluent and plasticizer: polyethylene glycol (PEG 400) •Dispersant: acidic polyether •Photoinitiator: TPO (diphenyl (2,4,6-trimethylbenzoyl)phosphine oxide) •Al_2O_3 size 0.88 μm (%) - 69.8	• Four Suspensions - M100, PEG25, O25, and O50	• PEG 400 increased cure depth, lateral growth, and surface roughness • While urethane acrylate led to decreased cure depth and lateral growth but increased surface roughness	[154]

samples were then analyzed to create a biomaterial with tailored mechanical properties. Adding PHB increases the young modulus, with a maximum value of 6.53 GPa achieved at 18 wt%, whereas the tensile strain decreases with 18 wt%. The resin mixture demonstrated the lowest tensile strain, measuring at 1.31 %. The crystalline nature of PHB affects the impact strength, leading to low energy-absorbing capacity and resistance to crack propagation and fracture toughness. Sun et al. [153] implemented a new stereolithography technique for printing hydrogels with adjustable mechanical properties and high resolution to overcome the challenges due to the low photo-reactivity of the high water content of hydrogels. They printed an acryloyl morpholine-acrylic acid-PEG diacrylate dry gel and underwent post-treatment with an aqueous solution that resulted in a soft hydrogel. The printed hydrogel allows for the separation of the printability of the precursor from the final properties. The resulting dual crosslinked hydrogel exhibits complex geometry, excellent stretchability and self-welding capabilities.

McAleer et al. [154] studied the role of resin composition in ceramic stereolithography and how they influence the printing parameters, including surface morphology and roughness, curing depth, layer height and lateral dimensions. They examined the role of the plasticizing agent polyethylene glycol - PEG25) in monomer-based suspensions. Four suspensions were examined; two had a 100 % monomer (ethoxylated trimethylolpropane triacrylate) suspension (M100) and a suspension with a 25 % by-weight plasticizing agent (polyethylene glycol) (PEG400). Whereas the other two had an oligomer (urethane methacrylate) added at 25 % (O25) and 50 % (O50) by weight. The plasticizing agent enhanced the curing depth, lateral growth and surface roughness, but the curing depth decreased in oligomers. Moreover, the oligomer exhibited increased surface roughness. This study illustrated how the different compositions of resin affect the printing parameters and provided insight into the design and optimization of materials. Some previous works related to the SLA process used for mechanical applications are reported in Table 2.

4.2. SLA for biomedical applications

The potential biomedical applications of stereolithography are numerous. However, one of its challenges is the absence of biocompatible and biodegradable resin for biomedical applications. Several studies are being conducted to introduce bio-resins that can be utilized in 3DP processes. These bio-materials are extracted from various sources and are divided into three categories, i.e. biobased, biocompatible and biodegradable [116]. Biobased materials are manufactured using biomass as a raw material, while biodegradable materials are manufactured from oil and are biocompatible. These polymers are either solid or soluble and are degraded in biological systems. The third type, biocompatible materials, must be compatible with living organisms without causing harmful effects [155]. Silbert et al. [156] synthesized methacrylate and acrylate resins derived from epoxidized sucrose soyate, a bio-based material. Three different resins were formed from

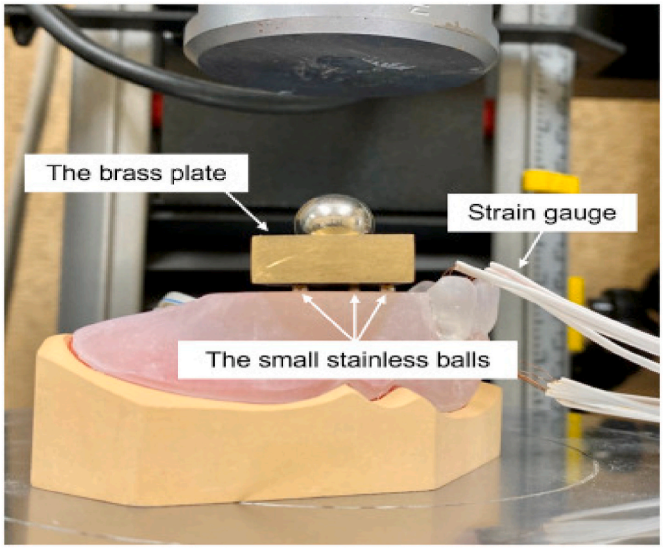


Fig. 11. Assessing the 3D printed dentures for distribution of stress. The weight applied was around 20N, at the rate of 20 N/s on a universal testing machine (Adapted from Ref. [159] Coprights © 2020 Elsevier Ltd. All rights reserved).

ESS: AESS (acrylate ESS), DMESS (di-methacrylate ESS), and MBSS (methacrylate butylated ESS). Specimens were printed at various orientations, and their rheological, tensile, flexural, and thermomechanical properties were evaluated and compared with those of commercially available SLA resins. This study highlighted the capability of modifying and incorporating biopolymers and affecting their properties. MBSS emerged as the most promising ESS derivative for flexural and tensile assessments due to its higher glass transition temperature (Tg) due to its methacrylate bonding and balanced cross-linking.

Meanwhile, Sedush et al. [157] introduced d,l-lactide oligomers biodegradable UV-curable, potential resin for SLA 3D printing. The resins were synthesized in linear and start-shaped structures with varying molecular weights, high temperatures, and cross-linking conditions. The material was a promising option for crafting biodegradable implants by SLA 3D-printing technologies without adding diluents. Some of the numerous applications SLA have by printing biocompatible and biodegradable material are described below.

4.2.1. *Patient-specific models and functional parts*

Additive manufacturing has been used excessively in healthcare, mainly in dentistry applications, because these processes provide customizable designs and patient-specific implants, dentures and other prosthetic devices [158]. Hada et al. [159] employed 3D CAD modelling and SLA processes to create a base for dentures. The dentures were 3D printed in three distinct directions of 0°, 45°, and 90°, and the influence of printing direction on stress distribution was investigated. It was concluded after experimentation that the lowest stress distribution occurs at a 45° angle. Fig. 11 represents the stress distribution on the printed dentures [159].

Addressing and mitigating its limitations and drawbacks is necessary to explore the full potential of additive manufacturing (AM) in dentistry. Krishna et al. [160] comprehensively analyzed dimensional accuracy, surface quality and mechanical properties of biocompatible materials for dental application through stereolithography printing. The test performed for quality assessment indicated that micro-surface roughness remains consistent at different inclinations, making it suitable for bone regeneration. The dimensional analysis results revealed negligible deviations between the CAD model and the printed structure, while the tensile test suggested the effect of sterilization on tensile properties. This study provides an overview of quality variations on AM processes and optimization of the processes to suit dental applications.

To bridge the gap between magnetic resonance imaging (MRI) analysis and additive manufacturing for creating patient-specific tissue substitutes, Kilian et al. [161] fabricated a model implant using two biomaterial inks, which mimics the relevant properties for bioprinting of the cells directly on the patient osteochondral defect. The geometric compatibility of the implant design was validated through the stereolithography model. This producer has the potential for broader application actions, various patients and different clinical scenarios, including defective tissues or organs.

4.2.2. *Implantable devices*

Aside from creating surgical models and functional parts, stereolithography can be used to develop patient-specific implants. These implants range from facial reconstruction in a trauma case by using the undamaged part image and forming a mirror image to using a CT image of a bone after removal of a tumour to craft support for the bone that degrades as the bone regenerates. Direct prototyping of implantable devices requires the resin to be biocompatible, biodegradable, and have precise appearance and mechanical properties. So, developing specific resins that fulfil the criteria is necessary for fabricating these devices.

Stereolithography prints various medical implants with specific geometries and properties, including components for artificial hips or knees, nerve guidance channels, prostheses and bone fracture devices [162]. Due to advancements in surgical reconstructive techniques, computer-aided design and manufacturing have also enabled the utilization of computerized tomography data for dental implants in surgery. Sudesh Dandekeri et al. [163] reviewed the stereolithography technique for creating surgical templates and its advantages over traditional manufacturing techniques. They concluded that identifying the bone anatomy before surgery facilitates maximizing the implant-bone

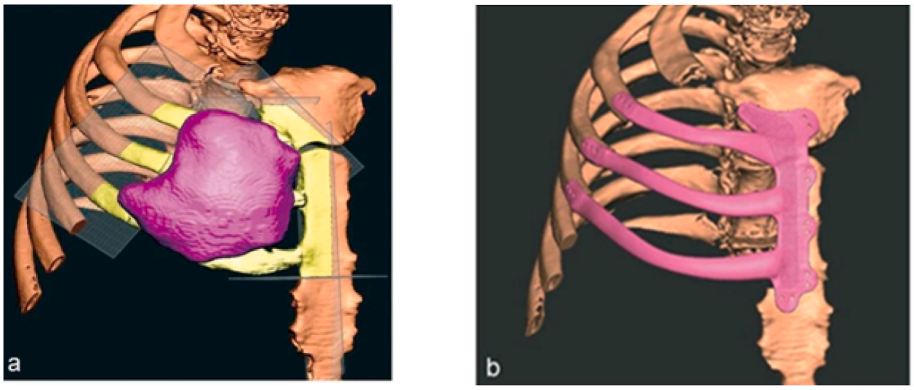


Fig. 12. (a) Digitally reconstruct the tumour from manual bone segmentation. (b) Subsequently, a titanium implant was customized to mimic the anatomical structure (Adapted from Ref. [165] Copyrights © 2020, The Author(s)).

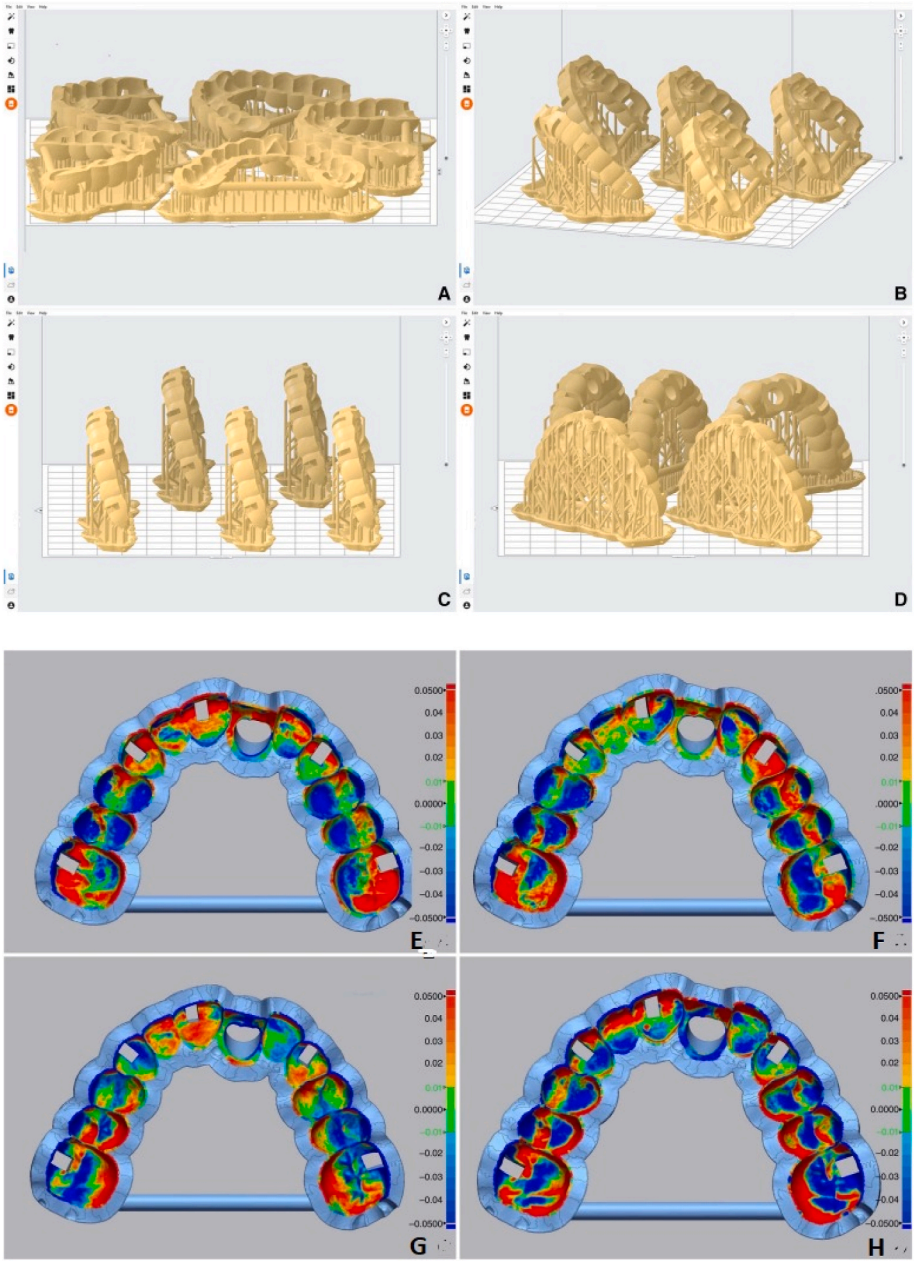


Fig. 13. Tested representative print orientations are depicted: A for Group 0, B for Group 45, C for Group 70, and D for Group 90. Each group tested shows a representative colour map discrepancy. The maximum range is 50 μm, the minimum range is −50 μm, and the tolerance is ±10 μm. E represents Group 0, F represents Group 45, G represents Group 70, and H represents Group 90. (Adapted from Ref. [166] Copyrights © 2023 by the Editorial Council of The Journal of Prosthetic Dentistry. All rights reserved.). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

interfaces and optimizes the results. This reduces operating time, the trauma of surgery, and recovery time, eliminating additional clinical or laboratory procedures.

Guillaume et al. [164] conducted a study of patient-specific implants (PSIs) for the orbital floor (OF) in the craniomaxillofacial (CMF) region made by stereolithography by employing biodegradable poly(trimethylene carbonate) (PTMC) infused with hydroxyapatite (Osteo-PTMC) while traditionally titanium meshes, were grafted. Guillaume performed orbital floor (OF) defect reconstruction surgeries on two groups of sheep and grafted Osteo-PTMC PSIs or pre-shaped titanium mesh implants in each group. Post-surgery, clinical CT, eye displacement analysis, implant stability and bone ingrowth assessments were performed regularly. Results indicated that Osteo-PTMC PSIs were superior to the traditional titanium mesh implants. This innovative approach provides significant advancement for OF defect reconstruction. Another study is based on the reconstruction of the chest wall after excision of the bone tumour [165]. The chest wall has a complex defect requiring skeletal and soft tissue reconstruction. Goldsmith et al. [165] and their team reconstructed the chest wall with custom-made 3D-printed titanium alloy ribs and hemi-sternum implants. Initially, the sternum and implant were 3D printed through the SLA technique, and the proposed resection of the sternum was simulated on the model. Upon evaluation, the ribs and hemi-sternum implant were found to be the precisely fit 3D model of the defect area, promoting the decision to proceed with the final implant. The final implant was manufactured using TiMG 1 powder bed fusion technology and designed by generating a 3-D virtual model of the patient's chest and the tumour from its CT scan data. The multiple-step approach of laser printing a custom-made anatomical titanium alloy chest wall implant yielded positive results while being patient-specific and preserving the thorax

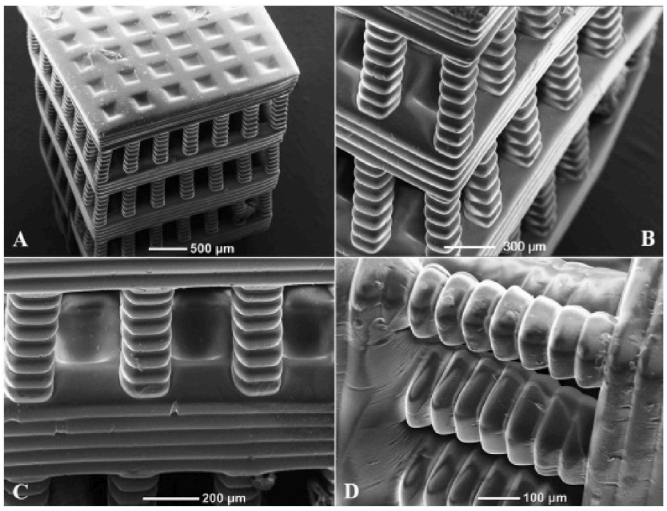


Fig. 14. Shows Scanning Electron Microscope images for the 3D printed lattice of 75:25 PLGA-diacrylate at (A) 27 times magnification, (B) 80-time magnification, (C) 100-time magnification, and (D) 150-time magnification. (Adapted from Ref. [173] Copyrights © 2020 Elsevier Ltd. All rights reserved.).

cavity structure, shape, and function. Fig. 12 shows the tumour and its placement [165].

To comprehend the influence of print orientation, storage conditions, and duration of storage on the precision of the printed implant, Salazar Rios et al. [166] used a biocompatible dental resin to fabricate four samples based on the print orientation: 0^0, 45^0, 70^0, and $90^{0\circ}$ and two sub-samples one being manufactured in light and one in dark, to check the storage effect after fourteen days. This in vitro study suggested that print orientation affects the precision of the implants printed with $0^{0,}$ showing the best manufacturing value. However, storage conditions or duration changes do not significantly impact the manufacturing accuracy values. Fig. 13 represents the discrepancy between the colour map of each group's print orientation [166].

4.2.3. Tissue engineering

Tissue engineering is rapidly developing in the biomedical field, where a scaffolding structure is incorporated with cells or biologically active compounds in the tissues to regenerate. Scaffolds are porous implants that offer temporary support structures for seeded cells and developing tissues. They act as templates for directing tissue formation and cell adhesion, requiring a well-defined internal pore network and external geometry. Scaffolds need to degrade into non-toxic compounds as newly formed tissue develops to ensure they do not cause inflammation after implantation.

Stereolithography has emerged as a standard method for fabricating biodegradable macromers and resins. Stereolithography was first used in 2000 to prepare biodegradable structures involving copolymers consisting of ε-caprolactone and trimethylene carbonate [167]. They used a ring-opening polymerization reaction and then modified the product with coumarin. Stereolithography has facilitated the fabrication of biodegradable scaffolds for tissue engineering from natural polymers with various adjustable properties [123,168]. Similarly, several studies have been conducted to explore materials suitable for 3D printing scaffolds by stereolithography. SLA requires a liquid photo-sensitive formulation that polymerizes upon illumination. Photo-initiated polymerization results in stable, non-degradable polymer chains, whereas degradation is necessary for the scaffolds so they do not accumulate in the body.

Similarly, Elomaa et al. [169] prepared a photo-crosslinkable poly (ε-caprolactone) (PCL)-based resin solvent free through stereolithography to construct tissue scaffolds. Methacrylation of three PCL oligomers with varying weights was prepared to generate highly gelled

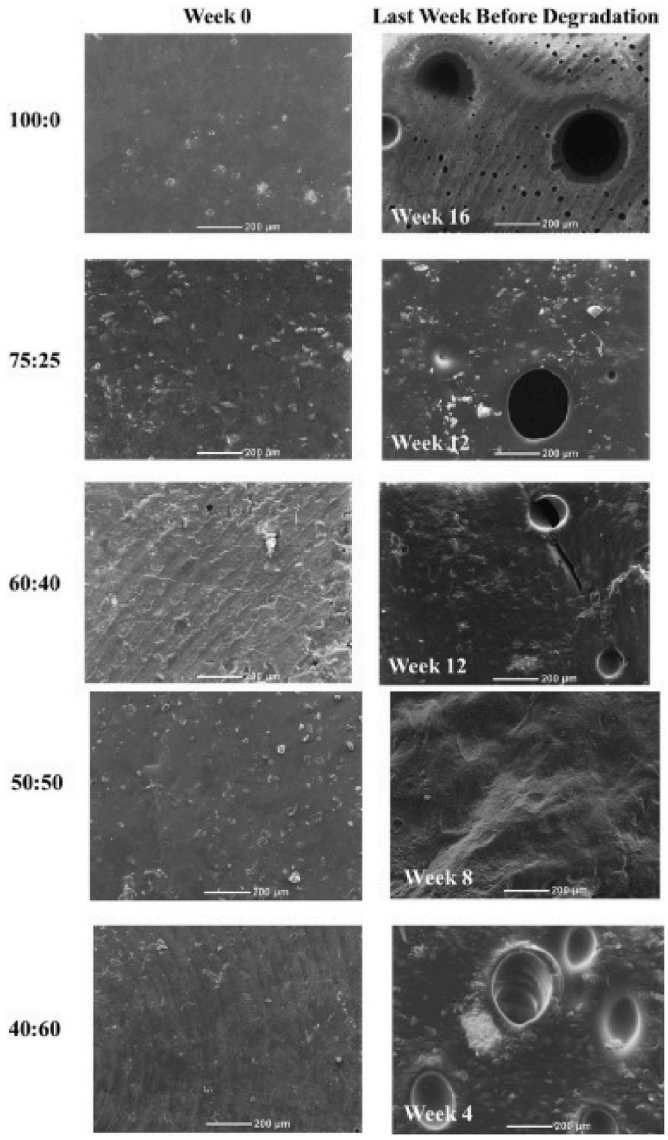


Fig. 15. SEM images illustrating the degradation at week 0 and during the final week on the film surface. (Adapted from Ref. [173] Copyrights © 2020 Elsevier Ltd. All rights reserved.).

networks by reacting with methacrylic anhydride. Porous scaffolds were fabricated through stereolithography by heating the resin to achieve the required viscosity. This also prevented the shrinking of the scaffold during printing, indicating great potential for the resin to be used in cell seeding and implantation. Chartrain et al. [170] by utilizing Poly(tri (ethylene glycol)adipate) dimethacrylate (PTEGA-DMA) fabricated biocompatible and biodegradable tissue scaffolds with 50 µm size through a mask projection micro stereolithography (MPµSL). They synthesized the resin and optimized the printing and curing parameters to fabricate the scaffold. With the addition of photoinitiator UV absorber avobenzone, they could develop tissue scaffolds smaller than 100 µm.

Ghalia et al. [171] conducted a review focusing on the impact of operating parameters, i.e. polymerization time, temperature, pressure, catalysts, and solvent polarity, on the synthesis (polymerization) process of polylactic acid (PLA). Optimizing the polymerization conditions can minimize side reactions and extend the molecular weight, which enhances thermal stability. The study examined the designs and production of porous PLA scaffolds with diverse porosities and pore sizes to emulate the intricate architecture essential for natural tissue regeneration. They also explored the diverse properties of PLA scaffolds

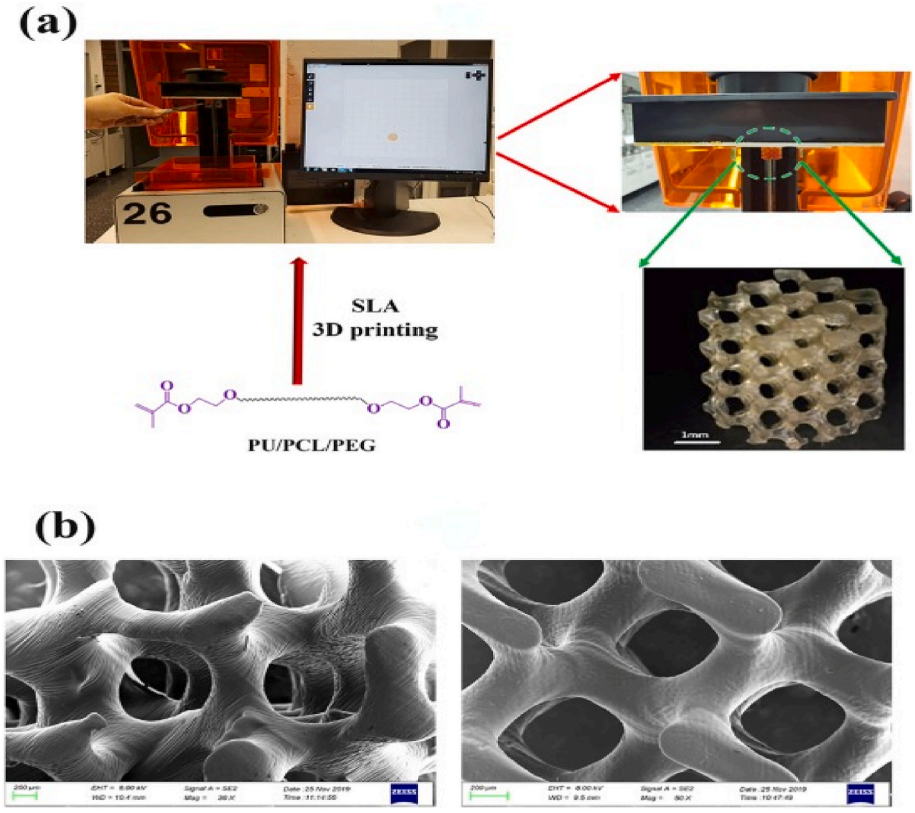


Fig. 16. (a) Manufacturing setup for fabricating 3D-printed scaffolds through SLA technique from photo-curable PU/PCL/PEG resin. (b) SEM images illustrating 3D-printed PU/PCL/PEG scaffold at 200 μm magnification (Adapted from Ref. [174] Copyrights © 2020 Elsevier Ltd. All rights reserved).

necessary for tissue regeneration. For more information on PLA scaffolds, kindly refer to Ref. [171]. Chandan et al. [172] focused their research on utilizing scaffolds for tissue regeneration of epithelial or mucosal cells. The process involved the design and three printings of the scaffold by employing a bio-compatible (Biomed amber) resin and printing it through a light-induced 3D printer. Through iterative processes, the parameters such as raft type, support density, touchpoint size, internal supports, layer thickness, and curing time were optimized and fine-tuned to balance the print quality resin usage, whereas the curing and washing enhanced its mechanical properties. Following computer analysis and particle tests, the scaffold printed in this study was deemed safe and displayed minimal deflection and stress values. The finding suggested it is suitable for medical use, indicating its feasibility for regenerating damaged skin cells and mucosal linings in anatomical regions such as the nasal canal, trachea, and oesophagus.

Commonly bio-degradable, photoactivated polymers require dissolution in non-reactive diluents, which can potentially cause anisotropic shrinkage and irregular part geometry in the final product. Wilts et al. [173] synthesized and characterized a solvent-free, photoactive polymer precursor with low viscosity, aiming for potential use in tissue engineering and addressing the challenges. Poly(lactide-co-glycolide) (PLGA) diacrylates (Fig. 14) oligomers with low number-average molecular weights (<1400 g mol^{-1}) were employed in vat photopolymerization. Upon cross-linking, these polymers exhibit mechanical strength and low cytotoxicity. Cytotoxicity and cell attachment studies indicated that the scaffolds facilitate cell growth and proliferation in soft tissue cells. Adjusting the ratio of lactide to glycolide with degradation rates between 4 and 12 weeks makes it suitable for many anatomical applications. Fig. 15 depicts the degradation through SEM imaging. This study also provided insight into the relationship between viscosity and its impact on mechanical and cellular properties.

Farzan et al. [174] utilize stereolithography to 3D print a polyurethane (PU) scaffold with biocompatible, degradable, and elastic properties for soft tissue regeneration. They employed poly(ε-caprolactone) (PCL) and/or polyethylene glycol (PEG) as microdiols and prepared three Pus resins, i.e. PU/PCL, PU/PEG and PU/PCL/PEG with the molar ratio of diol: HDI: HEMA was kept constant at 2:3:2 (Fig. 16). They investigated the influence of diols on hydrolytic degradation, thermal and mechanical properties, and hydrophilicity of polyurethanes (PUS). Further, they investigated the impact of various diols and scaffold geometries on toxicity and cell attachment. PU/PCL/PEG exhibited a higher tensile stress of 3.33 MPa and tensile strain of 145 %, surpassing the tensile strength of PU/PEG by 1.4 times and PU/PCL by two times. The printed PU/PCL/PEG scaffold did not exhibit cytotoxicity but had superior cell attachment and proliferation, making it a great candidate for soft tissue engineering, especially for scaffolds.

4.2.4. *Cell-containing hydrogels*

Cells are encapsulated into fabricated hydrogels to achieve high cell density and better control over cell distribution. Hydrogels are intricate three-dimensional networks formed by cross-linking hydrophilic polymers. Hydrogel contains water-soluble properties and hydrophobic components. They can attain their structures while retaining significant water [23]. Hydrogels can print mammalian cells while preserving their functionalities, precise drug delivery, and construction of living cells. Due to their biocompatibility and structural similarity to extracellular matrix (ECM), they are utilized in various fields, such as biomedicine, flexible sensors, and soft actuators [175]. Stereolithography is primarily chosen for hydrogel formulation due to its unique features, which include high resolution, precision, and customizability. These qualities ensure that hydrogel prepared by photocuring is suitable for various applications in biomedicine, tissue engineering, personalized devices, wearable sensors, and soft robotics [175].

Several researchers have employed di(meth)acrylated poly(ethylene glycol) (PEG-DMA) for the fabrication of hydrogels. Tikhonov et al. [27] formulated hydrogels based on polyethylene glycol diacrylate and

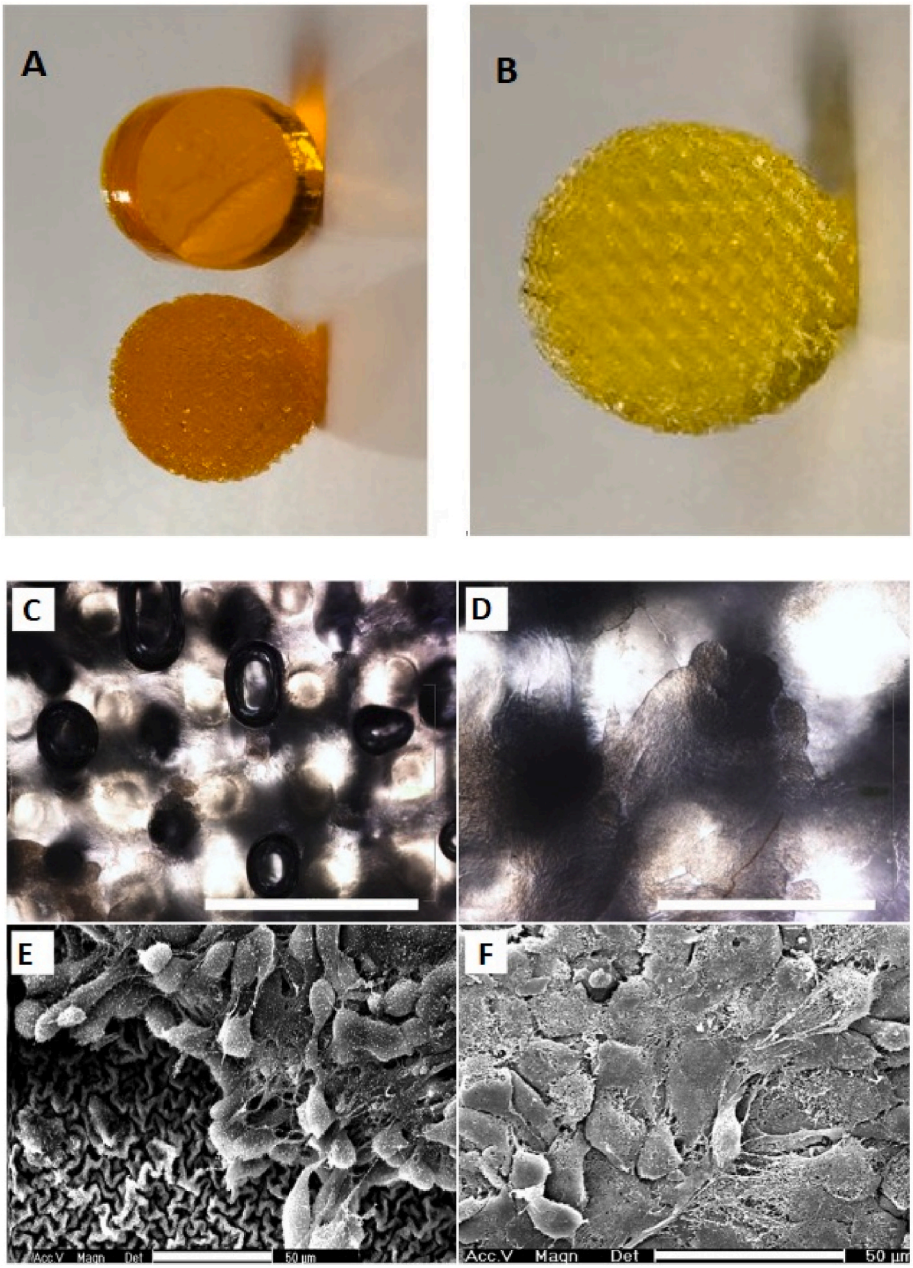


Fig. 17. (A) Hydrogel structure in Disk-shaped porous and solid shape Fabricated through SLA using an MA-PDLLA-PEG-PDLLA-MA based resin, (B) Gyroid Pore network design, (C, D) 24hrs later Light microscopy images and (E,F) SEM images after 24 h and after 5 days. (Adapted from Ref. [176] Copyrights © 2010 Elsevier B. V. All rights reserved.).

incorporated calcium phosphates (CaP, Ca/P < 1.5) to use the 3D prinited bio-composites for soft tissue regeneration. The structural characteristics and rheological and mechanical properties of the composites were assessed. The Young's moduli and mechanical strengths of the 3D PEGDA hydrogel composite scaffolds are augmented approximately 3–4 times when filled with octa-calcium phosphate. The viscoelastic properties of scaffolds revealed that CaP/PEGDA hydrogel composites are suitable for filling bone defects.

Seck et al. [176] studied the formulation of intricately designed three-dimensional porous and non-porous biodegradable poly(ethylene glycol)/poly(D, L-lactide) hydrogel structures, which were printed at high resolutions. The hydrogel structures underwent characterization and seeded with mesenchymal stem cells to assess their behaviour regarding cell adhesion and proliferation. The porous structure exhibited a defined pore network and connectivity among the pores, which is essential for cell proliferation and transportation of nutrients. It was concluded that the hydrogel structures are suitable for drug delivery, cell transplantation and other biomedical applications. Fig. 17 shows the hydrogel structures [176].

To know the processing-structure relationship in stereolithography 3D printing for hydrogels, Alketbi et al. [177] researched the effects of curing parameters and ink formulation on the 3D-printed PEGDA structures. They assessed the structural analysis, reactive coarse-grained molecular dynamics simulation, and microfabrication to explore the impact of photopolymerization on the structure's integrity. Their characterization impacted curing time, light intensity, and cross-linking degree on hydrogel elasticity and irreversible densification. It was concluded that hydrogels subjected to low-energy exposures and inadequate cross-linking experience irreversible densification during deswelling, which is attributed to insufficient network connectivity,

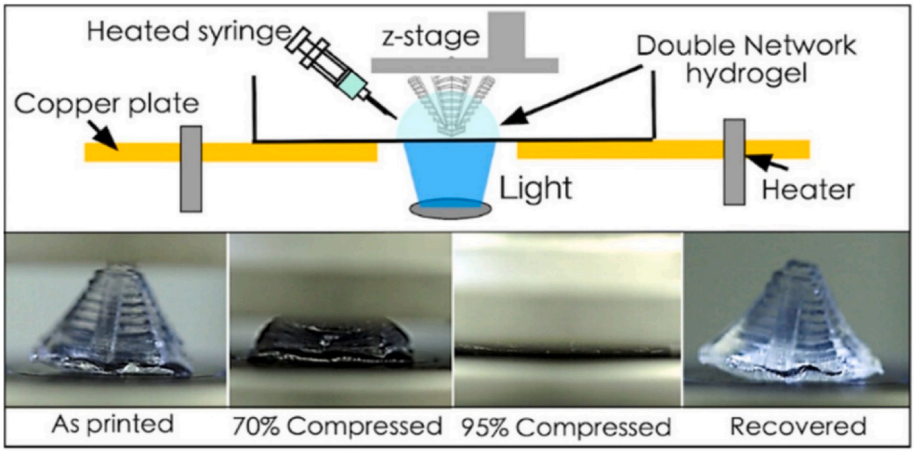


Fig. 18. Illustrating the novel technique for 3D printing of hydrogels with double-network for better mechanical properties. (Adapted from Ref. [179] Copyright © 2023 The Authors. Published by American Chemical Society. This publication is licensed under CC-BY 4.0.).

compressibility, and pores. Another similar study by Burke et al. [178] compared the Polyethyleneglycol dimethacrylate (PEGDMA) hydrogels prepared by UV chamber photopolymerization and SLA 3D printing. They successfully concluded that although there is no effect on thermal or chemical properties between the samples, the PEGDMA hydrogels printed through stereolithography exhibited increased wettability, hydrophobic properties and stiffness; this was attributed to heightened exposure to UV light during the SLA process. This research investigated the impact of UV light on hydrogel samples and their properties, presenting a potential avenue for adapting and modifying these properties for application in tissue engineering.

A novel method was introduced by Kunwar et al. [179] that shaped double-network (DN) hydrogels into customized 3D structures with superior mechanical properties. They utilized a one-pot projection stereolithography (PSLA) printing strategy that merged the properties of hydrogels (clearness and hydration) and PSLA (fast, high-resolution, and flexible 3D design) to design double network gels into intricate geometers with exceptional mechanical properties in tension and compression. The process begins with optimizing prepolymer formulation, compromising acrylamide and thermoreversible sol-gel κ-carrageenan into the first and second networks. Light-absorbing photoinitiator (LAP) is used to cross-link the polyacrylamide forming the first network while simultaneously κ-carrageenan undergoes a coil-to-helix transition leading to the interpenetration of the two networks and forming tough DN hydrogel structures. It was revealed that acrylamide (16 wt %), κ-carrageenan (2 wt %), MBAA (0.03 wt %), and photoinitiator LAP (0.12 wt %) made the optimal formulation. Tensil testing printed DN gel structures showed superior mechanical properties with higher fracture energy and modulus of elasticity. On the other hand, compression tests revealed excellent strain and stress capabilities. This novel technique can be utilized with various hydrogels to create innovative, smart, multifunctional devices suitable for various applications. Fig. 18 illustrates the novel technique for 3D printing of hydrogels with double-network for better mechanical properties. Table 3 summarizes some previous work on SLA for biomedical applications.

4.3. SLA for pharmaceutical applications

3D printing has a remarkable impact on the pharmaceutical sector, marking a significant milestone in the technological evolution of medicine. It has revolutionized pharmaceuticals and dosage forms and provides cost-effective means for medicine production. One of its key benefits is its ability to create customized dosage forms tailored to individual needs. The progress in technology has created new opportunities in the field of medicine, sparking ongoing research and exploration. 3D printing offers a versatile approach in pharmaceutical applications, from incorporating drugs to polymer solutions or directly within the printed item. SLA introduces the opportunity to manipulate drug release in the tablets by controlling in-fill density and surface area-to-volume ratio [185]. Wang et al. [186] conducted a study on the release of the drug paracetamol and 4-aminosalicylic acid from tablets of polyethylene glycol 300 fabricated through SLA. Similarly, Martinez et al. [187] investigated the impact of geometric shape on drug release by 3D printing tablets having paracetamol dispersed in PEG. It was concluded that surface to volume ratio affects the release of drugs.

Tropical or transdermal is another way to deliver drugs into the system. Drugs administered by this method are given either locally or systemically; hence, SLA-specific films and microneedles have been created to administer the drugs. Economidou et al. [188] used SLA to print patches of spear and pyramid-shaped microneedles to deliver insulin. Similarly, Choudhury et al. [189] employed SLA to create a polymeric film for delivering berberine in a controlled environment. Lu et al. [190] developed dacarbazine-loaded microneedles for extended drug release. This highlights the versatility of using stereolithography in different medical applications.

Due to conventional fabrication techniques limiting the complexity of microdevice structures, researchers are exploring microscale stereolithography to address challenges in oral drug delivery. Chang et al. [191] investigated the utilization of microscale stereolithography 3D printing to manufacture radiopaque micro-containers with optimized mucoadhesive geometry, aiming to improve bioavailability. They designed three distinct microcontainer types, i.e. neutral, pillar and arrow - each serving a specific purpose in drug delivery. Through optimization of printing parameters, they successfully fabricated these structures using stereolithography 3D printing. The studies using planar X-ray imaging showed variation in retention and transit time. Among the microcontainer designs, the arrow-designed microcontainer displays faster transit in the intestines, suggesting a great potential for targeted drug delivery. Overall, the research determines the potential of 3D-printed microdevices for oral drug delivery. However, further study is warranted to improve retention and transit time for efficiency in drug delivery. Fig. 19 demonstrates the microdevices 3D-printed radiopaque micro-containers with A) neutral, B) pillar, and C) arrow design, respectively.

4.4. SLA for electrical applications

Electrically conductive material utilization has expanded in various domains, including engineering, where they are employed to produce transducers, sensors, and actuators [6]. Piezoelectric ceramics are gaining interest due to their ability to convert compressive or tensile stresses into electric charges; however, the casting and machining

Table 3
SLA for biomedical applications.

Materials	Additives and Reinforcements	Printing Parameters/Material Concentration	Properties Improved	Ref
Three distinct Methacrylate and acrylate resins from (epoxidized sucrose soyate (ESS)- 41.99 % •AESS (acrylated ESS) •DMESS (dimethacrylated ESS) •MBSS (methacrylated-butylated ESS)	• Photoinitiator: (TPO-L, Ethyl (2,4,6-trimethylbenzoyl)-phenyl phosphinate) - 1.18 % • 1,6-Hexanediol diacrylate (HDDA) - 7.41 % • Diluent: BPAED - 49.41 %,	• Build Orientation - 0° and normal 90° • Post- Curing wavelength - 405 nm • Post- curing temperature - 50 °C • Sample Size - 76.8 mm × 13 mm x 4 mm	• Study highlights the feasibility of modifying bioderived materials with suitable reactive groups for SLA formulations. • MBSS - promising ESS derivative for flexural and tensile assessments, higher glass transition temperature (Tg) • DMESS - lower tensile strength and flexural behaviour, potential cracking due to excessive crosslinking • ESS resin – high viscosity, weak tensile and flexural behaviour	[156]
Methacrylate d,l-lactide oligomers	• Photoinitiator - Irgacure 2959	• Viscosity of the resin - 9–15 Pa s at 60°C, 0.5–6.2 Pa s at 100°C • Exposure time - 60 s • layer thickness - 3 mm	• Tensile mechanical tests revealed high values of Young's modulus 1–1.2 GPa (for star-shaped resin) and 1.8 GPa (for linear resin). • The resin at elevated temperatures exhibits rapid UV cross-linking rates and maintains low viscosity. • The best option for manufacturing personalized biodegradable implants with tunable properties	[157]
Methacrylate-based photopolymer resin	• PhotoInitiator: TPO Diphenyl (2,4,6-trimethylbenzoyl) phosphine oxide	• Thickness of layer – 100 µm • Wavelength – 405 nm • Three Print orientations - 0°, 45°, and 90° • Resolution along the x-y plane	• Changes in printing direction influenced the distribution of stress in SLA 3D printed dentures • Stress distribution at 45°was lowest which makes it more compatible.	[159]
Poly(ethylene glycol) diacrylate (PEGDA) molecular weights (250, 575, or 700 Da)	• Photoinitiators: Irgacure 819, TPO, or APi-180 • Orange or yellow dyes (E104 Quinoline Yellow) • CaP powder • CaP particles optimal with 50 wt % concentration	• PEGDA - 700 soluble in water • 1 wt% of Irgacure 819 relative to the monomer content was optimal • Concentration of E104 dye - 5 × 10^{-4} M	• An increase in monomer molecular weight resulted in a decrease in the absolute values of both elastic and viscous moduli (G' ≈ 20 kPa, G'' ≈ 2 kPa). • Increasing temperature decreases Viscous component and the elastic component • Composites (P-700_50/1_10%OCP) increase in the loss factor up to 0.1 Hz	[27]
Poly(trimethylene carbonate) (PTMC)	• Hydroxyapatite - 40 wt %	• PSI implant pores architecture: triply periodic minimal surfaces: IWP, diamond and gyroid) • Stiffness values for the original model were 0.270 N/mm, and cross-reinforced models were 0.472 N/mm. • Layer thickness - 50 µm • Irradiation times of 9 s per layer at 180 mW/dm2	• Osteo-PTMC implants are found to possess intrinsic osteoinductive properties and are biodegradable. • Osteo-PTMC shows promise as a synthetic bone graft substitute over titanium.	[164]
Biocompatible dental resin (Surgical Guide Resin V1)	Nil	• Four print orientation used - 0°, 45°, 75° and 90° • Thickness - 2 mm • Layer thickness −100 µm • Post wash - 20 min • Post curing - 24 °C for 30 min and ultraviolet (UV) unit for 30 min at 70 °C.	• The print orientation impacts the accuracy and precision of the inner surface of the implant. • The 0° has the highest value, whereas 90° has the lowest for accuracy and precision. • The storage conditions and durations have no significant impact	[166]
ε-caprolactone (CL) monomer **Three-armed hydroxyl-terminated poly (ε-caprolactone) (PCL-OH) oligomers –** molecular weights (Mn) 750, 1500, 2250, 3000 and 6000 g mol^{-1}	• calcium hydride (CaH_2) • Intiatior: tin 2-ethylhexanoate (stannous octoate $Sn(Oct)_2$) • co-initiator: trimethylolpropane (TMP) • Inhibitors: Hydroquinone and (±)-α-tocopherol (vitamin E) • Photoinitiator: Irgacure 2959 and Irgacure 369 • Dye - Orasol Orange G	• DMD Light - 1280 × 1024 pixels • Pore architecture – diamond and gyroid • Porosity - 70 % • Exposure time −10 s • Layer thickness - 25 µm • light intensity - 1600 mW dm^{-2}	• The PCL resin was successfully fabricated via solvent-free Stereolithography. • The scaffold pores showed high interconnectivity, indicating potential for applications such as cell seeding and implantation.	[169]
Tri(ethylene glycol) adipate Mn 1600 g mol^{-1} **Polyester, PTEGA-DMA**	• Dimethacrylate end-groups reaction of the PTEGA diol with 2-isocyanate ethyl365 Methacrylate • Avobenzone - 0.05 wt%.	• Laser Wavelength – 365 - 65 nm; • Light Intensity - 5.0 mJ/cm^2	• Incorporating a photoinitiator and the UV absorber tissue scaffolds with feature sizes smaller than 100 µm • Cell viability studies have shown excellent cell adhesion and significantly higher cell viability.	[170]
BioMed Amber	Nil	• Wavelength - 550 nm • Post Ccuring wash – 20 min	• The ultimate tensile strength of the printed scaffold was 73 MPa and within the elastic limit.	[172]

(*continued on next page*)

Table 3 (*continued*)

Materials	Additives and Reinforcements	Printing Parameters/Material Concentration	Properties Improved	Ref
		• Intensity of UV rays wavelength - 405 nm • Post- Curing temperature – 70^0 C for 30 min • Layer thickness −100 μm (0.1 mm) • Time - 19.865 s	• The study demonstrated the a bio-compatiblity and environmentally friendly scaffold fabrication for biomedical use. • The scaffold design and analysis indicated the potential for producing epithelial and mucosal scaffolds.	
Polycaprolactone diol (PCL, Mn 530 g mol^{-1}), Polyethylene glycol (PEG, Mn 400 g mol^{-1})	• Catalyst: dibutyltin dilaurate (DBTDL, 95 %) • Functionaizor: hydroxyethyl methacrylate (HEMA) • Dilutent: propylene carbonate • Dye: Orasol Orange G • Photoinitiator: ethyl phenyl (2,4,6-trimethyl benzoyl)phosphinate - 1.5 % wt	• Layer printed – 10 • Layer thickness - ~500 μm • The TGA thermograms of PU/PCL, PU/PEG, and PU/PCL/PEG were 258, 291, and 305 °C, respectively.	• The average molecular weight (Mw) for all resins was 6.7–9.3 kDa. • The number average molecular weight (Mn) ranged from 3.1 to 3.9 kDa • highest tensile stress (3.33 MPa) and tensile strain (145 %) was shoween by PU/PCL/PEG elastomer • Young's modulus was also high for PU/PCL/PEG elastomer, about 4.48 MPa • PU/PCL/PEG formed polymer gel in 3 s in curing • It has the potential to be used in soft tissue engineering	[174]
Methacrylate-functionalised poly (ethylene glycol)/poly(d,l-lactide) macromers (MA-DLLA-PEG-DLLA-MA macromer 33 wt% macromer)	• Water: 54 wt% • DMSO: 11 wt% • Photo-initiator: Lucirin TPO- 1.7 wt% • Phenol red: 0.19 wt% • Hydroquinone: 0.1 wt%	• Build platform step height of - 15 μm • Wavelength - 400–550 nm • Light intensities of - 20 mW/cm2 • The curing time per 15 μm thick layer was 32.5 s	•Fabrication of biocompatible hydrogel structure with tailored mechanical properties and porous architectures	[176]
PEGDMA monomers	• Photoinitiator: 1 wt% Irgacure 2959 or TPO	• UV curing system with an average intensity of 10–13.5 mW/cm2 • Layer thickness of 50 μm • Post-curing 10 min	• Increased stiffness values 13.01 MPa • Young's modulus 8.26 MPa • PEGDMA samples (30.6°) fabricated by SLA were were hydrophilic • UV chamber samples Tg values −42.5 °C, lower than the SLA samples −41.7 °C • Polymerization methods did not alter the chemical structure of the hydrogels • SLA 3D printed samples exhibited increased wettability	[178]
Acrylamide κ-carrageenan	• Photoinitiator: (LAP) - Lithium phenyl-2,4,6-trimethyl-benzoyl phosphinate	• Wavelength - 405 nm • Temperature - 80 °C • Resolution - lateral (37 μm) and vertical (180 μm)	• Tests demonstrated that the 3D design exhibited an ultimate stress and strain of 200 kPa and 2400 %. • Additionally, it displayed high compression stress of 15 MPa with a strain of 95 % when printed. • The printed structures exhibited high recovery rates.	[179]

ceramics are huge challenges. Hence, 3D printing or additive manufacturing is being explored as a potential manufacturing technique to achieve geometrical flexibility. Similarly, Chen et al. [180] showcased the feasibility of fabricating a piezoelectric-composite slurry containing BaTiO3 nanoparticles (100 nm) via a Mask-Image-Projection-based Stereolithography (MIP-SL) process. Through 3D printing of piezoelectric ceramics by utilizing high solid loading slurry with nanoparticles and post-processing of debinding and sintering. The printed parts exhibit a piezoelectric constant of 160 pCN and a relative permittivity of 1350. They could use the piezoelectric properties for biomedical imaging and other applications by integrating the printed parts into an ultrasonic transducer. The transducer achieves a 6.28 MHz ultrasonic scan, effectively illustrating the structure of a porcine eyeball. Fig. 20 illustrates (a) the optical image of the ceramic part fabricated by the MIP-SL system. (b) shows the profile of the transducer, and (c) demonstrates the optical image of ceramic in the transducer [180].

Due to the high demand for material with exceptional electromagnetic interference (EMI) shielding capabilities for application in aviation and vehicle manufacturing. Wang et al. [181] developed an epoxy-based composite micro sandwich consisting of solid and lattice layers via stereolithography. The structure assembly is arranged in a strategic architecture where the solid layer bears the load while the lattice structure inhibits rapid crack propagation. This architecture enhances strength by 92 %, increases toughness by 140 %, and improves fracture toughness by 18.5 %. The synergetic effect between the layers also promotes multiple reflections of the incident electromagnetic waves, resulting in increased EMI shielding.

The manufacturing of smart materials by additive manufacturing for scientific or industrial processes is rapidly growing. To enhance the electric conductivity of the materials, nanoparticles, metal powders, or composite polymers are incorporated. Similarly, Scordo et al. [182] selected poly(ethylene glycol) diacrylate (PEGDA) with the conductive filler poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) due to its biocompatibility and conductivity. They developed a highly conductive resin by dispersing PEDOT particles and optimized the composition by changing the photoinitiator, surfactant, and filler concentrations. The resin exhibited notable electrical conductivity (0.05 $S.cm^{-1}$) and mechanical properties (Young's modulus of 21 MPa). This polymer composite was found to have several potential applications, such as precision measurement, health monitoring, and sensing applications, such as microsensors, microfluidics devices, biosensors, and energy storage. Similarly, Kudo et al. [184] introduced the fabrication of Hierarchically Porous Carbon Microlattices (HPCMLs) using polymer composites through stereolithographic technology. Kudo and his team incorporated magnesium oxide nanoparticles (MgO NPs) and multilevel graphene sheets as UV scattering inhibitors. A simple cubic micro-lattice with 50 % porosity is designed, then carbonized at 1000 °C

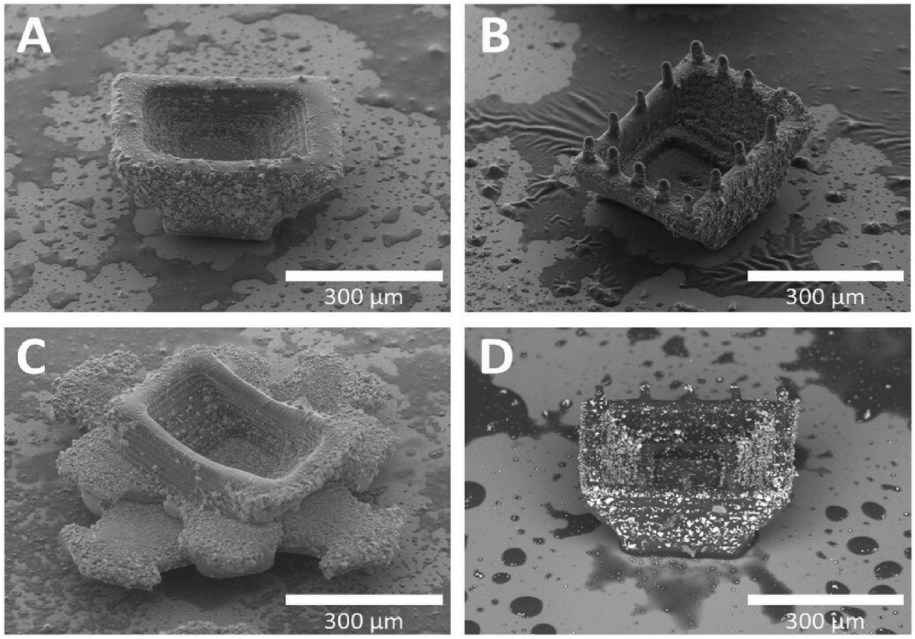


Fig. 19. Radiopaque micro-containers with (A) neutral, (B) pillar, and (C) arrow design while (D) SEM image indicating the BaSO4 particles in the pillar design (Adapted from Ref. [191] Copyrights © 2022 The Authors. Advanced Healthcare Materials published by Wiley-VCH GmbH).

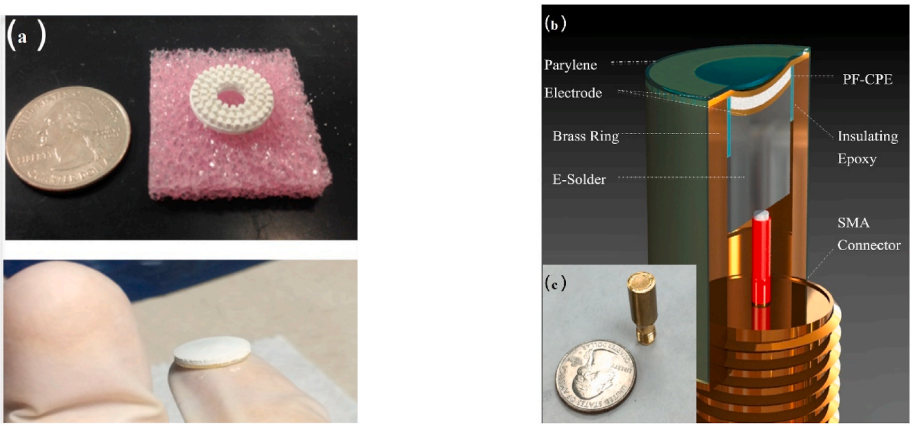


Fig. 20. (a) Demonstrates the optical image of ceramic part fabricated by MIP-SL system. (b) shows the profile of the transducer, and (c) demonstrates the optical image of ceramic in the transducer. (Adapted from Ref. [180] Copyrights © 2016 Elsevier Ltd. All rights reserved.).

in a vacuum, and chemical removal of MgO NPs is performed. The resulting microlattices exhibit hierarchical porosity with lattice architecture (≈100 μm), macropores (≈5 μm), mesopores (≈50 nm), and micropores (≈1 nm). During pyrolysis, the linear shrinkage reached up to 33 %, while the compressive strength of $\sigma = 7.45–10.45$ MPa and Young's modulus $E = 375–736$ MPa was achieved. These HPCMLs were used as bulk supercapacitor electrodes, with gravimetric capacitances of $C_{g,aq} = 84$ to 105 in aqueous electrolyte and $C_{g,og} = 12.7$ to 13.8 in organic electrolyte. They provide applications as energy storage devices and conversion and serve as bases for energy storage, sensing, and purification within structural components. Fig. 21 illustrates the four samples for each ratio in all the manufacturing steps, i.e., printed, pyrolyzed, and MgO removed [184].

Chen et al. [183] introduced a novel approach to providing highly porous ceramics using 3D printing technology. They showed the potential of employing the SLA technique to manufacture high-resolution ceramics with tailored properties. Porous alumina components were fabricated by mixing polyamide 12 (PA12) powder, alumina ceramic particles, acrylic resin as sacrificial porogen, aggregate, and mixture and printing through stereolithography. Incorporating aluminium sol enhances the mechanical strength of the porous alumina ceramics. The discrete element method model was utilized to simulate and elucidate the observed outcomes, offering insight into the optimization of the design method. Table 4 summarizes some previous work on SLA for electrical applications.

5. Challenges and future perspectives

Stereolithography techniques offer distinct advantages over traditional manufacturing techniques by enabling the manufacturing of detailed and intricate structures with exceptional precision, high resolution, rapid printing speed, and cost-effectiveness. Despite these advantages, certain challenges persist, forming the basis for future research initiatives. A comprehensive literature review reveals the following key insights into the realm of stereolithography. Fig. 22. Shows the challenges and future perspectives in stereolithography.

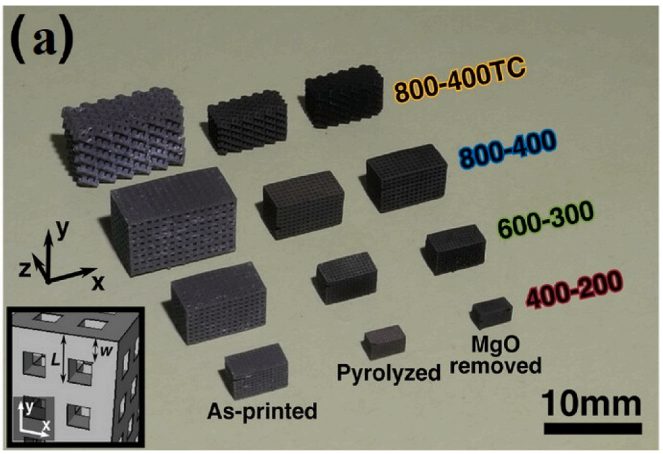


Fig. 21. Illustrates the four samples for each ratio in all the steps of manufacturing, i.e., printed, pyrolyzed, and MgO removed (Adapted from Ref. [184] Copyrights © 2023 Wiley-VCH GmbH).

- **Innovative Materials:** The material portfolio for stereolithography needs expansion as the existing photopolymers, such as acrylates or epoxy-based resins, are used, and there is a need to explore newer materials that are stronger, lightweight, durable, bio-compatible, biodegradable and sustainable. Additionally, natural materials should be incorporated into polymers in composites to enhance their properties and positively impact the environment. Exploring and testing multi-materials is also essential to investigate the advanced functional components. This will benefit the formation of complex structures and the customization of objects with different properties in different regions.
- **Uniform Distribution of Reinforcements:** Composites must have homogeneous dispersion of reinforcements/fillers in the resin matrix for optimum properties. Poor dispersion leads to the accumulation of the fillers, leading to an uncured photopolymer in the printed structure and apparatus, which leads to poor interfacial adhesion and agglomeration. Uniform dispersion of nanoparticles is necessary for consistent printing and electrical conductivity. So, ensuring uniform dispersion is crucial, and ways to overcome these challenges should be devised.
- **Production Cost:** Although AM processes are considered economical due to eliminating tooling costs, the post-processing requirements and energy consumption are not feasible and will cause economic challenges in mass production.
- **Optimization & Modeling:** A significant limitation of stereolithography is that the complexity of various printing parameters poses a challenge in predicting and optimizing the performance of the printed parts. There is an urgent need for advanced optimization and modelling tools capable of accurately predicting properties and optimizing functional parts tailored to specific applications.
- **Printing Parameters and Settings:** Certain mechanical properties, including tensile strength, young modulus, elastic modulus, flexural strength, elongation, and compressive strength, strongly depend on the printing parameters such as layer thickness, printing resolution, exposure time, light intensity, build orientation, light/laser penetration depth, curing time, post-curing time, sintering/debinding temperature, and size of reinforcing particles. Efforts should be made to reduce the linear shrinkage during curing or sintering processes so that the dimensions and mechanical properties are unaffected. Future aspects should focus on developing material with lower sintering temperature requirements to enhance the properties.
- **Standardized Tests and Assessment:** The development of standardized tests is necessary as a multitude of parameters are involved, affecting the process and the final product; hence, the tests would provide a fair evaluation of the differences.
- **Balancing Properties:** Balancing properties like mechanical and electrical is crucial as there is always a trade-off between them**.** High

Table 4
SLA for electrical applications.

Materials	Additives and Reinforcements	Printing Parameters/Material Concentration	Properties Improved	Ref
Photocurable resin SI500 Azeotropic mixture of methylethylketone (66 v/v%) and ethanol (34 v/v%)	• BaTiO3 powder (100 nm) 70 % • Dispersant (Triton x-100)	• Density - 5.64 g/cm3 • Dielectric constant - 1350 • Piezoelectric constant – 160 pCN	• Illustrated a structure of porcine eyeball. • Can create functional ceramic assemblies in 3D configurations	[180]
Epoxy Resin: EPON 826 (178–186 g)	• Curing Agent - Jeffamine D230 • Carbon Black (CB) – 40 nm, density 160 kg/m^3.	• Curing temperature - 60 °C • Curing time – 5 h	• Strength 92 % improved • Toughness 140 % improved • Fracture toughness 18.5 % increase • Shielding effectiveness 47 % increase	[181]
Poly(ethylene glycol) diacrylate (PEGDA)	• Filler - poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) • Photoinitiator - Irgacure 819 with weight 0.5 % wt., 1 % wt. and 6 % wt • Surfactant - Dimethylsulfoxide (DMSO) 5%wt	• Laser power density – 15 W mm−2 • Layer thickness (100 μm) • Scan velocity – 1000 mm/s •	• PEGDA: PEDOT (5:1) exhibited exceptional mechanical properties and electrical conductivity.	[182]
Curable Monomers: • Trimethylolpropane triacrylate (TMPTA) • Propoxylated neopentyl glycol di-acrylate (NPG2PODA) • 1,6-hexanediol diacrylate (HDDA)	• Plasticizer – TXIB • Photoinitiator - Irgacure 651 • Alumina (Al2O3) Ceramic Powder • Pore-forming Agent - Polyamide 12 (PA12) (diameter 58 μm) • Nano α-alumina Sol-gel	• Average pore size - 15 μm • Layer thickness - 125 μm • Shrinkage (7.58 % in x-y direction and 9.23 % in z direction) • Density - 2.04 g/cm^3	• Flexural strength increased from 37.89 to 51.36 MPa • Discrete Element Method (DEM) was utilized to simulate the mechanism.	[183]
Epoxy-phenol photocurable resin	• Magnesium oxide nanoparticles (MgO NPs) diameter 0.05 μm • Graphene nanosheets (size 25 μm)	• Four different mass ratios tested - (100:5:1, 100:10:1, 100:15:1, and 100:20:1) • Optimum was 100:15:1 • Pyrolyzed 400 °C and elevated till 1000 °C • To remove the MgO NPs – soaked in HCL at 60 °C for 1.5 day and then dired at 120 °C.	• HPCMLs can be employed in structural energy storage, sensing, and purification, integrating both structural and functional properties	[184]



Fig. 22. The challenges and future perspectives in stereolithography.

loading of filler material and conductive polymers leads to poor mechanical properties and strength due to their brittle nature.

- **Conductivity compared to metals:** The ability of conductive polymers and nanocomposites to compete with metal components in specific applications is challenging despite advancements in achieving high electrical conductivities. Bridging the gap and demonstrating the advantages of conductive polymers is essential to foster widespread adoption.

6. Conclusions

This paper's objective was to review the additive manufacturing technique, Stereolithography, which constructs 3D objects by solidifying the rein through light-induced photopolymerization. This paper dives into the development of the stereolithography technique since its inception and is categorized into four generations in chronological order. From scanning to projection, SLA cures each layer while building a structure. Whereas the introduction of DMD causes high resolution and production rate, continuous stereolithography boosts the printing speed. The volumetric process introduces an innovative concept, which enables it to create intricate structures in a single operation. The paper extensively explores the chemistry of the resin during the photopolymerization process. It thoroughly discusses all the resin components needed for 3D printing, highlighting techniques to ensure a homogeneous resin mixture to ensure perfect structures. This paper proceeds to elucidate current applications of the stereolithography process and explore strategies to enhance its positive effects. Furthermore, a detailed discussion about the challenges and prospects is presented, providing a comprehensive insight into the evolving terrain of stereolithography.

Due to its continuous development and innovation, Stereolithography has found widespread applications in various fields. However, this process has shortcomings, such as achieving high resolution, larger building size, and rapid printing speed. The material limitation is also a significant challenge, as developing specific processes and systems tailored to materials with desired properties is necessary. While additive manufacturing techniques are relatively in their early stages of development, numerous research opportunities have yet to be explored. Nevertheless, the continuous development and innovation in AM manufacturing techniques indicate a promising future with sustainable outcomes.

CRediT authorship contribution statement

Ambreen Afridi: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ans Al Rashid:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Muammer Koç:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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